Supplementary Evidence: Evaluating LLM and Agent Side-Channel Defenses

This supplement provides the search accounting, corpus maps, saved empirical results, dual-coding records and source-bound audits accompanying the main paper. The package README maps these sections to machine-readable data and the read-only verifier. Original failed and negative outcomes are preserved alongside positive results.

S1. Search and Evidence Audit

TABLE S1. HISTORICAL ROUTE ACCOUNTING FOR THE ORIGINAL OCTOBER 8, 2026 SNAPSHOT, INCLUDING ITS LATER SOURCE CHECKS. COUNTS OVERLAP AND DO NOT FORM A COMPLETE PRISMA FLOW; THE OCTOBER 21 SUPPLEMENTARY ROUND IS REPORTED SEPARATELY.

Route / disposition	Count
Broad IEEE Q4 hits; not fully screened	35,743
Amended Q4b export; recall not verified	847
IEEE saved export occurrences / distinct records	2,079 / 2,001
IEEE full-text referrals / background routes	25 / 205
ACM saved occurrences / publication groups	16,687 / 13,201
ACM Oct. 14 provisional exclusions / full-text referrals / unresolved	9,368 / 883 / 2,950
ACM dispute groups / combined check candidates (Oct. 18)	332 / 57
57-item source check: scoped dispositions / unresolved	57 / 0
Backward reference occurrences (40 fixed seeds)	2,237
Forward seeds resolved / distinct returned works	38 / 24
Recent finite reconciliation: overlapping decision rows	109
Source-limited candidates not admitted / background chapter	4 / 1
Snapshot core / adjacent studies	43 / 17
Snapshot coded study families	60

The 57 scoped dispositions are not completed full readings. B069's complete official abstract supports background routing; later-page eligibility remains unchecked. The 109 rows overlap; 883/2,950 are dated global routes. Q4b does not replace unscreened Q4. This snapshot's coded-set cutoff is October 8, 2026; its counts do not describe the later expanded collection.

Targeted discovery and seed-anchored snowballing. Because native IEEE/ACM keyword exports (Table S2) captured only 1/24 and 0/24 of the 24 core seed papers, our primary acquisition combined venue-targeted proceedings screening (USENIX Security, NDSS, IEEE S&P, CCS, PMLR), targeted arXiv discovery, and one-hop backward (2,237 reference occurrences) and forward (24 distinct works across 38/40 resolved seeds) citation tracing from 40 fixed seeds, followed by the completed 565-record October 21 Semantic Scholar and native arXiv search frame (Section S10).

Historical frame accounting and independent dual-coding. Every prepared metadata ID in the historical frame has an explicit disposition (4,753 unique IDs: 2,778 exclusions, 116 referrals, 1,859 unresolved routes; Table S1). While historical exclusion checks (1,300 checks in the 1,555-record frame; 23-case AI-on-AI exercise with 17/21 determinate agreement; 332-dispute readbacks) used the single AI-assisted workflow, the revised package supplements them with independent human/expert dual-coding across 100 stratified screening records ($\kappa = 0.886$), 144 split-audit cells ($\kappa = 0.876$), 24 retrospective transfer cases ($\kappa = 0.866$) and 150 pairwise comparability judgments ($\kappa = 0.916$; `experiments/revision-m1-m8/`).

S2. Family Index, Corpus Maps and Prior Work

The accompanying *Supplementary Evidence* archive is the access route for the full tables and fit-free checks. Its README starts with the current paper, family index, result ledger, utility/cost inventory and claim links. Each result joins a citation-key family, source version/hash and locator to its receiver/intervention fields and supported judgment; Table S4 is a filled example. Separate case-evidence entries identify additional source subconditions, rather than silently pooling them into the 33 scoped inventory records.

The merged index enumerates 82 families: 58 core (20 agent-native, 38 runtime/application-inherited) and 24 adjacent, including five authorized diagnostic components. The original 60-family index is a separately labeled historical snapshot. Mixed semantic attribution remains adjacent under the auxiliary-only rule. The package preserves the snapshot's 33 scoped result records/66 field positions and the complete original empirical and workflow analyses. S10 supplies the 22 added-family rows and expansion sensitivity. Source locators and hashes support inspection; third-party full texts/raw traces are obtained from their named sources.

Roles of the remaining indexed sources. Additional core studies concern content/hidden-property inference [2], [3], [4], [5], [6], [7], identity/provenance attribution [8], [9], [10], [11], and authorized integrity diagnostics [12], [13], [14]. Adjacent audit, execution, browser-data and profiling studies supply boundary comparisons rather than additional auxiliary privacy attacks [15], [16], [17], [18], [19], [20],

TABLE S2. EXECUTED NATIVE QUERIES AND SAVED EXPORT OCCURRENCES. IEEE DATES: OCTOBER 8, 2026; ACM QUERY/PARTITION ACQUISITION: OCTOBER 8–9, 2026. COUNTS ARE NOT INCLUDED STUDY FAMILIES.

ID	Exact query	IEEE saved/hits	ACM saved/original hits
Q1	`("large language model" OR LLM) AND ("side channel" OR "side-channel" OR "timing attack")`	63/63	479/479
Q2	`(agent OR agentic) AND ("traffic analysis" OR fingerprinting OR "metadata leakage" OR "side channel" OR "side-channel")`	671/671	6,181/6,181
Q3	`("KV cache" OR "prompt caching" OR "speculative decoding") AND (privacy OR leakage OR attack)`	37/37	511/511
Q4	`("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage)`	0/35,743	Not run
Q4b	`(("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage) AND ("large language model" OR LLM))`	847/847	4,026/4,026
Q5	`("language model" OR LLM) AND ("traffic shaping" OR "cache isolation" OR padding OR "side-channel defense")`	95/95	3,689/3,688
Q6	`("language model" OR chatbot OR "AI assistant" OR "generative AI") AND ("side channel" OR "side-channel" OR "traffic analysis" OR fingerprinting OR "timing leakage")`	366/366	1,801/1,801

IEEE used Command Search/default metadata, without a lower-year or document-type filter; every result page was exported as RIS with Citation and Abstract. Q4's zero means not exported in the October 8 acquisition, not zero hits or a completed screen. Q4b supplements Q4; recall preservation is unverified. The later Q4 prefix does not reconstruct historical membership. ACM used the All field in the ACM Full-Text Collection (not Guide), then native E-Publication Date partitions and All Results/BibTeX export. Partition receipts retain exact dates and URLs. Q4b's June bin has 285 displayed versus 284 saved; agreement of the overall saved/original counts does not repair the missing identity. Q5's partition total differs from its original whole-query count. Final-named export files were byte/hash matched on October 9; these two retrieval discrepancies remain. The October 8 coded-set cutoff is checked separately from acquisition partitions.

[21], [22]. The index records each receiver, target relation and evidence scope; the reference list includes these indexed sources, not just the central case papers.

S2.1. RQ sensitivity

The original expansion adds procedural skills, physical receivers, topology shifts and architecture timing to RQ1–2. The latest finite search additionally changes the positive defense evidence: CachePrune and OTRO supply useful mechanism-bound privacy/quality/performance conditions. RQ3 still requires generator-specific bridges; RQ4 separates simulation, component overhead, fixed guardrails and executed protection. S10 reports these changes explicitly. Section S6 retains the original source-deletion analysis; the new controlled experiments independently instantiate its first two placement mechanisms.

S3. Retained Failure-Inclusive Controls

The 288-trace fixed-policy pilot has call-count-determined labels and saturated coarse F1=1.00, so it supports no sequence-increment claim. In the separate tool-front-end study, the observer sees both tools; the constrained baseline succeeds on 19/24 tasks, below the frozen promotion gate. These local results are not passive-network or deployed-defense evidence.

The retail case retains 148 windows, 49 empty observations and 102 stopped-error episodes. All three frozen loss contrasts have intervals including zero; test goal achievement is 7/40 versus 8/40 and joint utility remains unresolved. Neither this null-compatible estimate nor failed tasks establish absence of leakage or effective defense. Full protocols and results remain in the delivered evidence package.

S4. Public-Trace Reanalysis: Protocol and Boundaries

UI Traces (commit `4f3df18`) supplies 4,200 records/14 declared identities, retaining 1,806 error/absent-result indicators. Labels are declarations, not authenticated backends or task correctness. The supplemental package binds source bytes, code and receipts.

Predictors use JS types, local time and callback order, excluding collector time/navigation, URL/DOM text, questions, answers and diagnostics. Cross-document order assumes an archive receiver, not authenticated website capture. Exact-question-disjoint train/validation/test groups have 2,043/1,022/1,135 rows and 395/202/226 questions (823 distinct questions). The source split identifies FRAMES but has no semantic task-family field. Full identity–dataset/exact-question support and a seven-bin first-word *grammatical proxy*, explicitly not a semantic family, are supplemental; neither semantic independence nor family balance is established.

The original 19-fit comparison selects weighted logistic/forest candidates on validation loss only; selected models are forests. H2 adds type bigrams to coarse counts/page-age. Original intervals use 3,000 paired-question resamples and a nominal 98.333% three-contrast allocation, conditional on fixed fitting/selection and noise realizations. Archived H1's identical selected predictor yields exactly zero difference

TABLE S3. ITEM-BY-ITEM RECONCILIATION OF THE 565-RECORD OCTOBER 21 SEARCH FRAME (BEFORE AND AFTER THE 204-RECORD EXCLUSION AUDIT) AND THE TRANSITION FROM THE 60-FAMILY SNAPSHOT TO ALL 82 CODED FAMILIES.

Route / family category	Pre-audit	Audit Δ	Final (Indexed)
Excluded: no hidden C2/C3 target	221	−4	217 (0)
Excluded: outside LLM / diff. target	175	0	175 (0)
Existing snapshot core / adjacent	23 / 1	+1/0	24 / 1 (in 60)
New source-verified core families	14	+1	15 (**+15**)
Adjacent source-reviewed contexts	28	+2	30 (**+7**)
Adjacent metadata-only contexts	28	0	28 (0)
Background / proposal metadata	58	0	58 (0)
Missing abstract / unavailable text	17	0	17 (0)
565-frame total / Net added families	**565**	**0 net**	**565 (+22)**
Coded collection: Core / Adjacent (**Total**)	Snapshot: 43 / 17 (**60**) → Full: 58 / 24 (**82**)		

The exclusion audit selected 177 high-risk pattern matches plus 27 reason-stratified records (seed 20261021), rather than all 396 initially excluded records. This enriched 204-record subset rerouted 4 records out of the 221 initial C2/C3 exclusions: (1) `SUP-0192` (MIRAGE [1]) moved to new core ($14 + 1 = 15$, all 15 added to the family index); (2) `SUP-0552` (InputSnatch arXiv alias) merged into existing family `inputsnatch2024` ($23 + 1 = 24$, +0 new families); and (3–4) `SUP-0191` and `SUP-0424` moved to source-reviewed adjacent contexts ($28 + 2 = 30$, of which 7 boundary-defining studies enter the adjacent index and 23 remain in the 565-record ledger). Thus $60 + 15$ (core) $+ 7$ (adjacent) $= 82$.

and interval $[0, 0]$; it is a selection illustration, not equivalence evidence.

For Q questions with n_q records each, loss is $Q^{-1} \sum_q n_q^{-1} \sum_{i \in q} -\log p_{i,y_i}$; accuracy substitutes the correct-argmax indicator. Fit weights $N/(Qn_q)$ equalize question weight. Add $\epsilon = 10^{-6}$ to every class probability, renormalize, then clip true-label probability to $[\epsilon, 1]$. Bootstrap draws Q questions with replacement; original contrasts share draw indices, with percentile endpoints 0.8333% and 99.1667%.

Controls declared with original outcomes known. The additive protocol is frozen before new fits, with original outcomes known. All-14 common support is empty. Select the largest identity subset with at least 20 common training questions, then greatest support and lexical tie-break: Gemini 3.1, Gemma 4-31B, Seed 2-lite, with 32/10/10 questions. Hash order selects one row per question/identity, retaining failures; test has 30 rows. This is a three-class sensitivity result.

The first-Wikipedia-document *proxy* stops at HTTP navigation, beforeunload, unmarked origin/URL change or clock reset, but not SPA history alone. Markers gate membership and never enter features. No frame/document IDs authenticate membership; late callbacks may cross boundaries.

TABLE S4. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	`dangol2026workflow` / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [23].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about 13× bytes and 11× latency in that condition.
Unresolved fields	Objective terminal task utility NR-R; precise paired live-refit comparison NR-V.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

Empty views remain and epochs are not pooled. The supplement gives the complete six-event projection example and boundary implementation.

Three row-seeded shuffles preserve counts/timestamps/coarse features in *all partitions*, refitting combined candidates with coarse retained. Fits total 8 matched-task, 8 document-proxy, 8 clean and 12 shuffled; all 36 finish before test prediction, without convergence warnings. Clean H2 reproduces exactly. Each shuffle selects coarse: zero increment is then an algebraic consequence of selecting the same predictor. This checks the selection pipeline, not whether shuffled sequence features contain no information; a forced-sequence comparison would be a different, unperformed analysis.

New intervals use 3,000 question resamples, seed 20261018 and 98.333% allocation across the three new contrasts, conditional on fits and shuffle streams. They are exploratory, not blind confirmation or global adjustment across all historical analyses. Separate code reconstructs points/intervals from saved probabilities. Matched-task uncertainty and the 509/1,135 test error flags (3.4–91.5% by identity) prevent intrinsic-identity claims.

The saved-result verifier supplies local replay; it does not establish independent replication or live utility/cost.

Failure/activity diagnostics. The configuration was frozen with prior outcomes known: three representations, four candidates each, original weights, seed 20261008; all 12 fits precede test prediction. Failure-only uses error/absent-result bits; empty-only uses no projected events; activity-only uses log-count plus seven page-age summaries, not cross-document duration. Candidate hyperparameters and projections are supplemental. No new confirmatory intervals are supplied. Flags cover 43.00% overall and 44.85% test rows, not authenticated task success. Stratifying 509 flagged/203 questions versus 626 unflagged/216 questions changes support; increments are not a causal failure effect.

TABLE S5. PRIMARY OBSERVABLE/RECEIVER PROFILE × SIGNAL ORIGIN ACROSS ALL 58 CORE FAMILIES (SHOWING SNAPSHOT 43 → FULL 58). COLUMN n COUNTS DISTINCT FAMILIES; EACH FAMILY OCCUPIES ONE PRIMARY RECEIVER PROFILE ROW BUT MAY SPAN MULTIPLE GENERATING ORIGINS (P1–P8).

Observable / receiving permission	n (43→58)	P1	P2	P4	P5	P6	P7	P8	Representative sources
Flow / destination / routing metadata	$12 \rightarrow 16$	0	10	1	7	2	2	3	AgentPrint; WRA; NetEcho; LLM-WFIN; Routing Metadata
Own-site actions / browser / harness	$5 \rightarrow 8$	0	0	1	8	2	6	0	UI Traces; FP-Agent; WebAgents;AgentProv; Lidar
API duration / cache / split-return	$12 \rightarrow 14$	0	14	1	0	0	0	0	Early Bird; LeakyLMs; CachePrune; SplitDecoy
Resource / physical / sparsity / page	$10 \rightarrow 15$	15	6	0	1	0	0	0	TDXRay; OTRO; PAGE404; SparLeak; SparSEEty; PhysicalEdge
Trajectory / procedure / diagnostic log	$4 \rightarrow 5$	0	1	1	5	2	0	0	SIGLEAK; ClawGuard; cycle detection; MIRAGE (diagnostics)
Total distinct core families	$43 \rightarrow \mathbf{58}$	Each family contributes once to n (20 workflow-origin A, 38 inherited-runtime/application B).							

P1 hardware; P2 serving; P4 application/API; P5 orchestration; P6 tools; P7 environment; P8 multi-agent. P3 denotes transport (coded as a receiving projection). Five Type IV authorized diagnostic components (ClawGuard, cycle detection, LumiMAS, LLM RCA, MIRAGE) are separated from Type I confidentiality claims.

TABLE S6. FROM PRIOR CROSS-LAYER SYSTEMATIZATIONS TO RESULT-LEVEL DEPLOYMENT DECISIONS: THE SPECIFIC COMPARISON OBJECT AND ITS DEMONSTRATED USE.

Prior work	Established perspective	Added result-level question	Demonstrating analysis
Banerjee et al. [24]	Cross-layer compound AI attacks can relax threat assumptions; MITRE organization	Does cover end at the protected interface, get reserialized, or execute as an action?	NetEcho final-hop check and Dangol/AHB transport/action distinction; explicit conditions for permitting transfer.
Forough et al. [25]	Confidential-computing threat model; partial inference-defense transfer; isolation is not side-channel immunity	Which oracle, execution mode and component-cost result licenses the proposed defense transfer?	Early Bird KV/semantic split and KVGov hardware/simulation/analytic separation.
Cai et al. [26]	Traffic features, shaping objectives and bandwidth trade-offs	Can a traffic-cover cost/utility argument be used for a dummy tool action?	Dangol/AHB: a state-machine boundary determines which task/policy outcomes must additionally be measured.
Debenedetti et al. [27]	Auxiliary ML components induce privacy channels	Distinguish an execution-mediated oracle from ordinary semantic attribution	C3 boundary and mixed-output eligibility judgments.
Lahjouji–Colaco [28]	Agent data surfaces, governance, composition and cross-session risks	Which bridge test would license a specific transfer?	Receiver/intervention and generator/holdout result maps.
Kim; Horváth et al. [29], [30]	Broad agent security; physical DNN mechanisms/threat models	Compare source and receiving privileges without a universal strength order	Remote calibration versus own-site, local and privileged diagnostic profiles.

Semantic-family balance remains unavailable.

Exact H3 perturbation and adaptation. The unchanged perturbation sets $t'_{ire} = t_{ie} + u_{ire}$, $u_{ire} \sim U(0, 5000\,\mathrm{ms})$, independently per valid page-local timestamp in source-array order. Missing times consume no draw; types/order/page keys remain fixed. Three row-seeded realizations transform all partitions. Each adaptation clones the clean-selected representation/architecture and fits once on perturbed training rows with identical weights, without retuning or additional data (three fits). Both programs predict the same perturbed test rows; average paired losses across realizations, rows within question, then questions. The supplemental exact protocol supplies entry points, seed strings and invariants. The transformation edits stored times; it does not execute action delays.

S5. Intervention Inventory and Evidence Dispositions

The companion mode-specific table and 33-record ledger distinguish executed inference, offline/simulated evidence and component costs. All 66 utility/cost positions retain the source, scope, state and consequence for a named claim or inventory-only role. The main deployment-question table does not treat these heterogeneous records as a common success/failure denominator.

Four textual defense assertions lack the quantitative binding needed for a matched effect (Table S13). Their unbound quantities are excluded from comparison; their qualifying attack evidence remains. AHB has verified condition-wise refit code, but historical D1-run and paired D2 are recorded as NR-V, not reconstructed. Its task outcome is not inferred from that code.

D1 denotes protected-condition fitting/calibration; D2

TABLE S7. SOURCE–RECEIVER–INTERVENTION MAP. THE TARGET COLUMN DISTINGUISHES CONTENT/PROPERTY INFERENCE, ATTRIBUTION AND HUMAN–AGENT DISTINCTION; NONE ALONE ESTABLISHES AN UNAUTHORIZED PRIVACY VIOLATION. U DENOTES AN UNRESOLVED SOURCE/RUN BINDING.

Source / target	Signal origin → receiver	Intervention / evaluated mode	Surviving path or restriction	Comparison consequence
UI Traces / identity attribution	Agent actions → own-site JS	Recorded timestamp delays; offline D2	Types and order preserved	Test residual action features; no live task-cost estimate.
NetEcho / content inference	Serving → network A/B traces	Upstream padding/batching; protected-data fit	Decoded upstream content/state can be reserialized downstream	Inspect final-hop emission; defended fitting alone does not establish a matched adaptation contrast.
SIGLEAK / skill nondisclosure	Procedure → task/output receiver	Common prompt defense	Procedural signatures remain	Compare same receiver/target; not a paired installation effect.
AHB / human–agent distinction	Touch behavior → app/detector	Executed fake touches; separate transforms	Task state may change; fit binding U	Measure transitions and objective utility; do not import offline delay costs.
WRA / private-intent inference	Real/decoy research requests → visited domains	Concurrent execution; interleaved-domain evaluation	Intent reconstruction falls with one decoy	Preserve domain-view privacy benefit; terminal decoy utility and cost are separately unreported.
Dangol / workflow-property inference	Protocol/traffic → defined link	Stored captures versus live OHTTP (§9.7, OHTTP paragraph after Table 5)	Feature ablation differs from network masking	Keep accuracy 0.17 capture and 0.30 live populations separate.
Early Bird / cache state	Cache reuse → API timing	KV simulation versus GPT-Cache prototype	Distinct matching mechanisms	6ms/4% applies to semantic prototype only.
KVGov / private prompt	Shared cache → probe oracle	Simulated inference; real timing; analytic isolation	Hash-key isolation not semantic-index isolation	Hardware gap does not become deployed defended ASR.
FP-Agent / identity attribution	Policy/browser → visited site	Task/site evaluation, mixed attributes	Static configuration may remain	Task holdout is not independent-visitor holdout.
Traffic benchmark / topology	Multi-agent workflow → selected links	Explicit topology/view shift	Unseen generators differ from hidden links	Cross topology with receiving-view restriction.
LeakyLMs / architecture	Inference → streaming client	Chosen-prompt timing; no measured defense	Same-GPU-class calibration for architecture	API reachability is not hardware-independent recovery.

TABLE S8. DEPENDENCY–HOLDOUT–CLAIM MAP. EACH ROW EXCLUDES ONE EXPLANATION, NOT ALL CONFOUNDING.

Dependency / source	Required holdout	Claim licensed; residual limit
Task / UI Traces	Question/task-family	New task units; fixed harness remains.
Browser / FP-Agent	Visitor/configuration	Configuration transfer; task-only split does not test it.
Workflow / traffic benchmark	Agent topology or composition	Tested topology shift; no all-orchestrator guarantee.
Hardware / LeakyLMs	GPU/serving calibration	Tested architectural transfer given calibration; not unknown-hardware recovery.
Repository / coding fingerprints	Repository and generation pathway	Tested attribution domain; not covert human-mediated use.
Order / worked case	Same units, refit shuffle	Fixed-program order increment; not task-independent identity.

TABLE S9. FIXED-ARCHIVE PREDICTIVE EVIDENCE AND MATCHED SUPPORT. UNITS: NATS OF QUESTION-WEIGHTED LOSS REDUCTION. POSITIVE VALUES FAVOR THE SECOND PROGRAM. INTERVALS CONDITION ON FITTED PROGRAMS; THE ORIGINAL AND ADDITIVE-CONTROL ALLOCATIONS REMAIN SEPARATE.

Contrast	Estimate	98.333% interval
Worked case: coarse vs. combined (H2)	0.3447	[0.2974, 0.3944]
Matched questions, 3 identities[a]	0.2634	[−0.2584, 0.7241]
Reanalysis: clean vs. adapted (H3)[b]	0.1503	[0.1205, 0.1815]

[a] 30 rows/10 test questions, 3 classes, rather than 1,135 rows/226 questions, 14 classes; not a direct effect-size comparison. Its interval uses the additive three-control allocation. Archived identical-predictor H1 remains in the protocol.
[b] Both programs predict the same timestamp-perturbed test rows; three fixed realizations are averaged.
Cross-document order assumes an archive receiver, not verified ordinary-website capture. Intervals vary test-question weights, not training, task families or deployments. No live defense utility or cost is inferred.

denotes a paired training-procedure comparison on the same defended test condition. Neither entails feasible labeled-data access or matched budgets. Token payload, plaintext padding, wire bytes, model loading, map lookup and added service latency remain distinct cost objects. A scoped NR-R completes the requested extraction, rather than asserting zero cost or preserved utility.

TABLE S10. PUBLIC-TRACE PERMISSION PROJECTION. "SITE" MEANS ITS OWN DOCUMENT ONLY; IT DOES NOT AUTHENTICATE CROSS-DOCUMENT CAPTURE. FULL FIELD-LEVEL CSV AND ONE COMPLETE INGESTED-RECORD PROJECTION ARE SUPPLEMENTAL.

Field	Site JS	Analysis use
Type, page-local time	Yes	Original predictors; activity diagnostic
Within-document order	Yes	Type bigrams
Cross-document order	Unverified	Archive-receiver bigrams
URL/origin	Own page	Origin gate only; no text/page-key feature
HTTP navigation, collector time	No	Boundary proxy only / time excluded
Error/result metadata	No	Analyst-only failure baseline
Task/answer metadata	No	Excluded; question hash splits/weights only
DOM/link text	Own page	Excluded from all predictors
Document/frame ID	Absent	No authenticated document membership

TABLE S11. POST-REVIEW H2 CONTROLS. POSITIVE VALUES DENOTE A QUESTION-MEAN LOG-LOSS REDUCTION (NATS); BRACKETS ARE CONDITIONAL 98.333% RESAMPLING INTERVALS. NOT A NEW AGENT RUN.

Contrast	Test rows / questions	Effect [interval]
Matched tasks, 3 identities: coarse minus sequence	30/10	0.2634 [−0.2584,0.7241]
First-document-epoch *proxy*: coarse minus sequence	1135/226	0.1504 [0.1193,0.1825]
Original clean minus mean of 3 refit shuffle increments	1135/226	0.3447 [0.2961,0.3953]

All 14-class common support is empty. The first-document view is reconstructed, not authenticated. Its 0.1504 feature increment and H3's 0.1503 timestamp-adaptation effect are distinct contrasts on the same archive, not independent replications. Cross-document order assumes an archive receiver. All three shuffles select coarse, so their zero increments make this contrast's 0.3447 point equal to original H2. The intervals use separate frozen bootstrap seeds (20261018 here; 20261008 for original H2), yielding different finite-resampling endpoints. Zero selected increments do not establish information equivalence. No task utility or live defense is measured.

S6. Conclusion Sensitivity

Table S15 exposes direct-source dependence in the original snapshot. Within that snapshot, removing NetEcho withdraws the only downstream re-emission instance; removing AHB withdraws the measured mobile state-change instance, while WRA's privacy-oriented decoy result remains. Removing WRA withdraws that private-intent decoy result while AHB retains the task-state branch. UI Traces and SIGLEAK remain different surviving-feature instances, not replications. Removing Dangol withdraws its capture/live contrast; Early Bird, KVGov and SpliceDefense retain separate component examples. S9/S10 separately report the later controlled evidence and source expansion; they are not part of this historical ablation denominator.

Starting from the original 60/43/17 total/core/adjacent families, deleting AHB, WildAuth and coding fingerprints leaves 57/42/15; deleting its four diagnostics leaves 56/39/17; joint removal leaves 53/38/15. These are snapshot ablations, not the later 82-family collection. The snapshot's four diagnostic/semantic-boundary additions are not premises of the cover/action contrast. For its five multifeature-trace studies, unresolved bindings bound D1 at three–five and D2 at two–five; removing AHB gives three–four and two–four. Early Bird calibration is a separate stage.

Excluding all four unbound defense quantities leaves the three deployment cases unchanged, because none uses those quantities as a premise. AHB's detector-run unknowns prevent an adaptive-detector claim but do not erase its separately reported task-success points. Its non-Fake online point is an additional Table 1 subcondition in the case ledger, not an extra independent experiment or an imputed value in the combined-condition inventory row. Whole-service costs, training access and missing utility admit no favorable imputation or ranking.

Narrowing H2 to matched tasks leaves task-independent identity unresolved; the document proxy does not authenticate capture. H3 supports only same-record offline adaptation. The original joint workflow endpoint remains negative, including the missing-score bounds and post-outcome sensitivity views. These positive, inconclusive, zero and negative outcomes are preserved with their original endpoint

TABLE S12. ADDITIVE FAILURE/ACTIVITY DIAGNOSTICS ON ALL 1,135 TEST ROWS/226 QUESTIONS. POINT SCORES USE THE ORIGINAL QUESTION WEIGHTS; NO NEW CONFIRMATORY INTERVALS. ORIGINAL PROGRAMS ARE READ FROM SAVED PROBABILITIES, NOT REFITTED.

Program	Loss (nats)	Accuracy (%)
Training prior (saved)	2.64267	6.59
Error/result flags only[a]	2.47464	14.25
Projected-empty flag only	2.63832	6.79
Activity/page-age only[b]	1.81732	37.68
Original coarse (saved)	1.44058	51.29
Original coarse+sequence (saved)	1.09591	66.26

[a] Analyst-only diagnostic, not an observer-valid attacker; two binary flags, not error text. [b] Total event count and seven local-page-time summaries; no action types/order, diagnostics, text or session duration. Same four candidate architectures and validation-only selection for each of the three new representations (12 fits). Low-dimensional baselines do not rule out higher-order failure/task confounding.

TABLE S13. QUANTITATIVE ASSERTIONS EXCLUDED FOR MISSING BINDINGS.

Source	Missing binding for quantitative use
AgentPrint	Referenced supplementary mitigation result, procedure and population not located.
Weiss	No matched vendor-defense before/after outcome located; the artifact appendix reproduces an attack table.
Gu	No provider-change result with a matching comparison population located.
PromptEcho	MSS/threshold padding is described; exact threshold, population, cost denominator and defended outcome are unbound.

Sources: [31], [32], [33], [34]. No missing value is imputed; withholding these assertions does not remove the four families.

TABLE S14. SCOPED INTERVENTION DISPOSITIONS ACROSS ALL 58 CORE FAMILIES IN THE FULL COLLECTION (ALONGSIDE THE 43-CORE OCTOBER 8 SNAPSHOT AND 15 NEW CORE ADDITIONS). CATEGORIES ARE MUTUALLY EXCLUSIVE AT THE FAMILY LEVEL.

Source-scoped disposition	Snapshot (43)	Added (+15)	Full (58)
Evaluated mitigation component/condition	16	+4	20
Fixed common guardrail (not paired installation)	2	+0	2
Other intervention / decoy boundary / objective	4	+1	5
No evaluation identified in inspected scope	13	+9	22
Type IV diagnostic component (confidentiality NA)	4	+1	5
Unbound assertions excluded from effect comparison	4	+0	4
Total core families (Result records)	43 (33)	+15 (+6)	58 (39)

The four October 21 evaluated-mitigation additions (`wu2026cacheprune`, `lee2026otro`, `chen2026sparleak`, `hadad2026shape`) contribute six source-bound result records (bringing the result ledger from 33 to 39 records across 78 utility/cost fields); `politis2026splitdecoy` evaluates forward split-model decoys against backward gradient-support leakage, and `revankar2026mirage` is the fifth Type IV diagnostic component.

meanings.

S7. Retained Indexing Experiment and Scenario Illustration

The retained model-assisted experiment compares flat checklist R, structured checklist T and raw SCOPE traversal V. There are 512 roots per arm and three query judgments per root (1,536 judgments). The endpoint is the mean of each root's three binary joint outcomes, requiring a correct deployment decision and exact configuration counts. R has 624 joint successes (40.63%); V has 494 (32.16%). T has 693 observed successes and three missing queries belonging to one root, giving the retained 45.12–45.31% missing-outcome range. V is worse on this joint endpoint; this is retained negative evidence.

Decision-correct but count-incorrect judgments number 537 (R), 476 (T) and 695 (V), out of 1,536 query opportunities. Observed queries have no schema/evidence-ID failures. Binary-decision successes are 1,161 (R), 1,169 (T, plus three missing) and 1,189 (V). The 1.11–1.30 percentage-point V–T binary difference is a secondary post-outcome decomposition, not proof of a better workflow. Counts are query judgments, not independent roots; percentages average three judgments per root. We do not infer causality from the error decomposition.

The separate 24-scenario exercise uses author-constructed source-reuse cases and author labels. Seven are Compatible, twelve Incompatible and five Unresolved. Its two supplied coder columns match 23/24 and 22/24 labels and agree on 21/24. The original headline rule is always Compatible (7/24); always Incompatible gives 12/24 and always Unresolved 5/24. No accuracy advantage over an informative, equal-information comparator is claimed. All three-category confusion matrices are in `experiments/reviewer-audit/scenario_confusions.json`. Independent construction, blinding and out-of-corpus validation remain unperformed.

S8. Result-Pair Comparability, Nine-Item Split Audit and Dual-Coding

This audit applies the paper's comparison rules to the 33 result records across 22 families (and the nine-item split reporting checklist to the 16 baseline and 4 newly added evaluated-mitigation families). The supplied second-coder columns are retained for agreement analysis; coder independence and blinding are not verified by these files (`experiments/revision-m1-m8/`).

S8.1. Within-family and within-cohort pairwise comparability

While the machine-readable inventory retains all $\binom{33}{2} = 528$ unordered pairs for completeness, 426 pairs cross unrelated mechanism cohorts and are outside scope by definition. Table S17 summarizes two nested informative subsets (the 13 are included in the 102): the **13 within-family pairs** and the **102 within-cohort related pairs** (traffic-envelope: 66; cache-service: 28; behavioral humanization: 3; generator fingerprint: 3; local-memory: 1; fixed guardrail: 1).

Within the 13 same-family pairs, seven compare aligned outcomes and six substitute different primitives or cost objects. Of the seven aligned-outcome pairs, four have positively different collection or execution populations (such as Dangol's stored-capture feature ablation at 0.17 accuracy versus live loopback OHTTP at 0.30 accuracy) and three lack a verified paired-test binding; none licenses direct cross-record subtraction. Across the 50 dual-coded pairs (all 13 same-family pairs plus 37 stratified related-cohort pairs = 150 judgments across descriptive, paired-effect and transfer rules), recorded two-column agreement is **94.0% (141/150)** with **Cohen's** $\kappa = 0.916$ (95% bootstrap CI $[0.856, 0.963]$; `m7_dual_coding_pairs.csv`).

S8.2. Nine-item split reporting audit and supplied coding records

Table S18 uses R4 for defended fitting, R5 for a matched adaptation contrast, R7 for inference, R8 for utility and R9 for cost across both the 16 original evaluated-mitigation families (144 cells) and the 20 total evaluated-mitigation families (including `wu2026cacheprune`, `lee2026otro`, `chen2026sparleak` and `hadad2026shape`; 180 cells).

TABLE S15. LOAD-BEARING SOURCE ABLATION WITHIN THE ORIGINAL 60-FAMILY SNAPSHOT. ENTRIES IDENTIFY ITS ACTUAL DEPENDENCE; S9/S10 REPORT SUBSEQUENT CONTROLLED EVIDENCE AND SOURCE EXPANSION SEPARATELY.

Conclusion / source locator	What is supported	What is lost on source removal / NR dependence
Action preservation / UI Traces 6.1,Fig5	Recorded delays preserve structure and allow adaptation	Action-delay instance and one D1/D2 positive lost; exact release-point binding not needed for protocol claim.
Procedural preservation / SIGLEAK 4/C	Skill inference persists under common extraction defense	Skill-defense instance lost; no paired installation-effect claim.
Re-emission / NetEcho IV-C–E	Downstream reserialization can restore fine granularity	Only direct downstream re-emission instance lost; no multisource replication claimed.
State change / AHB 5.2.3/Table1/E.3	Fake actions can cause unintended UI transitions	Only executed mobile state-change instance lost; detector-run binding remains U.
Private-intent cover / WRA 8.1/Table8	Real/decoy execution with interleaved-domain inference evaluation lowers one-decoy E_{func}	Private-intent decoy instance lost; AHB state-effect instance remains. Same-condition terminal utility/cost are NR-R.
Mode split / Dangol §9.7, OHTTP paragraph after Table 5; 9.8–9.9/Table6	Capture frontier differs from live protocol and timing ablation	Concrete accuracy 0.17/0.30 pair and one D1/D2 positive lost.
Cost boundary / Early Bird V-A/B; Splice VII-C; Speculation 6.1; KVGov 5.9	Separate component and deployment costs	Removing one withdraws its example, not other boundary instances; full service costs remain unknown.
Generator dependence / UI Traces, FP-Agent, traffic benchmark, LeakyLMs	Task, browser, topology and calibration tests differ	Each source's axis withdrawn when removed; no pooled portability estimate.

TABLE S16. AUTHOR-CONSTRUCTED SCENARIO EXERCISE: COMPLETE CONFUSION MATRICES. EACH TRIPLE GIVES PREDICTIONS (COMPATIBLE, INCOMPATIBLE, UNRESOLVED) FOR THE INDICATED TRUE CATEGORY. THE LAST TWO COLUMNS DISTINGUISH DETECTING MISMATCHES FROM RETAINING COMPATIBLE REUSE.

Procedure	True C	True I	True U	I recall	C recall
Always C	7,0,0	12,0,0	5,0,0	0.0	100.0
Always I	0,7,0	0,12,0	0,5,0	100.0	0.0
Always U	0,0,7	0,0,12	0,0,5	0.0	0.0
Coder column 1	7,0,0	0,12,0	0,1,4	100.0	100.0
Coder column 2	7,0,0	0,11,1	0,1,4	91.7	100.0

This illustrates the recorded labeling exercise; scenarios and labels were not independently held out. C recall is specificity when Incompatible is the positive class and evaluation is restricted to the C/I subset. Unresolved is retained as a third outcome.

TABLE S17. DESCRIPTIVE COMPARABILITY WITHIN THE 13 SAME-FAMILY PAIRS AND 102 RELATED-COHORT PAIRS. SUPPLIED TWO-COLUMN CODING OVER 50 PAIRS (150 JUDGMENTS) REACHED 94.0% AGREEMENT ($\kappa = 0.916$).

Subset	Compatible	Incompatible	Unresolved
Same family ($n = 13$)	7 (53.8%)	6 (46.2%)	0
Related cohort ($n = 102$)	16 (15.7%)	80 (78.4%)	6 (5.9%)

Across the 144 baseline cells, supplied two-column coding (`m7_dual_coding_checks.csv`) reached **93.1% raw agreement (134/144)** and **Cohen's** $\kappa = 0.876$ (95% bootstrap CI $[0.798, 0.943]$). All 10 pre-adjudication differences occurred across adjacent *Partial* versus *Bound*/NR-R/NR-V boundaries (e.g., whether AHB's verified refit code without Table 1 run provenance is *Partial* or NR-V on R4, and whether WRA's separate Table 11 visibility utility is *Partial* or NR-R on R8). Full cell-by-cell rationales and source locators are preserved in `m6_split_checks.csv` and `m7_dual_coding_checks.csv`.

TABLE S18. NINE-ITEM SPLIT REPORTING AUDIT OVER THE 16 ORIGINAL EVALUATED-MITIGATION FAMILIES (AND 20 TOTAL INCLUDING THE FOUR OCTOBER 21 MITIGATION ADDITIONS IN PARENTHESES). R1–R9 DESCRIBE EVIDENCE FOR SCOPED REUSE CLAIMS; THEY ARE SEPARATE FROM DEPLOYMENT CHECKS 1–3.

Audit item	Bound	Partial	NR-R	NR-V	NA
R1. Target & protection polarity	16 (20)	0 (0)	0 (0)	0 (0)	0 (0)
R2. Receiver fields & linkage	13 (17)	3 (3)	0 (0)	0 (0)	0 (0)
R3. Placement & action dispatch	13 (16)	2 (3)	0 (0)	1 (1)	0 (0)
R4. Protected-condition fit	4 (4)	1 (1)	0 (0)	4 (6)	7 (9)
R5. Matched adapted test	2 (2)	2 (2)	0 (0)	5 (7)	7 (9)
R6. Workloads & generator shifts	14 (18)	2 (2)	0 (0)	0 (0)	0 (0)
R7. Defended inference effect	14 (17)	0 (1)	2 (2)	0 (0)	0 (0)
R8. Task or component utility	2 (3)	2 (3)	10 (11)	0 (0)	2 (3)
R9. Mechanism or protocol cost	10 (11)	1 (3)	4 (5)	0 (0)	1 (1)

Bound: explicitly established for the scoped claim. Partial: a supported element coexists with a proxy or unlinked operating point (e.g., AHB refit code without verified Table 1 run binding; WRA Table 11 visibility utility separate from Tables 8–9 decoys). Only one family across the expanded 20-family mitigation set (`wu2026cacheprune`) jointly binds R7, R8 and R9 on the same operating point. Supplied two-column agreement over the 144 baseline cells is 93.1% ($\kappa = 0.876$, 95% CI $[0.798, 0.943]$).

The current `combined_result_records.json` makes the six expansion records explicit alongside the 33 preserved baseline records: 39 records from 26 distinct families, while the reporting audit covers 20 mitigation families. The six additions are recordizations of the supplied source-bound extraction, not a new independent coding pass.

S9. Controlled Replay Details and Revised Accounting

The current experimental source is `experiments/reviewer-audit/recompute.py`, with the instru-

mented retail implementation in `retail_replay.py`. Historical inputs and previous summaries are retained unchanged; revised JSON files take precedence for the current manuscript. No fresh model generations were requested.

TLS corpus and length accounting. The 48 saved outputs comprise 24 brief/expanded pairs. Training and testing each contain 12 topics and 24 outputs. T1 uses the original verbosity labels. T2 retains brief responses and takes the first two period-delimited sentences of expanded responses. T3 uses the first two and first three sentences of each expanded response. T4 retains both original outputs per topic and labels their domain. There is one generation per prompt, not repeated model sampling. T4 has 14 positive and 10 negative test outputs (seven versus five topics), so 50% is not its empirical majority-class baseline.

For every regime and framing policy, `tls_records.json` saves topic, split, label, plaintext bytes, application-record count, payload bytes, wire bytes and recovered plaintext hash. `tls_gradient.json` reports absolute means and matched denominators. All 48 texts are used for cost means; only 24 held-out outputs are used for accuracy. A record contains five TLS header bytes plus 24 encryption-overhead bytes under the pinned TLS 1.2 suite. Handshake traffic, TCP/IP and timing are outside this cost scope. Bucket 1024 uses four 256-byte writes per quantum, hence 1,140 rather than 1,082 wire bytes per quantum. Buckets 256/512 use 128-byte writes.

The training-fitted cap rounds the maximum framed training length, 1,194 bytes, up to 1,280 bytes. It emits ten records of 128 bytes (1,570 wire bytes). Overflow adds 128-byte frames, never truncates, and leaks an overflow-dependent length; none of the saved test responses overflows. The loose 16-record, 1,024-byte-per-record control is 16,848 wire bytes and is not a lower bound on protection cost.

Cohen's absolute standardized mean difference, recomputed across the 48 constructed plaintext lengths using pooled sample variance, is 12.264 (T1), 5.723 (T2), 3.418 (T3) and 0.033 (T4). Previous values embedded in regime labels were not produced by that calculation and are withdrawn. These descriptive values are not entropy, independent tests or a model of required framing cost.

Adaptive fitting and bootstrap hierarchy. The clean threshold is fitted to training TLS payload bytes. The defended thresholds use the first four or all twelve training topic pairs. The 1024B defended threshold is 1,680 TLS payload bytes. Stochastic padding has 50 draws; the first is measured on sockets and 49 are calculated using socket-checked record accounting. The independent resampling unit is one of 12 test topics. For each topic, correctness or paired gain is averaged across its two outputs and all draws before 10,000 percentile-bootstrap resamples (seed 20261021). We do not refit the model in that bootstrap or infer uncertainty across fresh model generations. Stored per-topic/per-draw correctness permits exact reconstruction. Both constant caps give a constant specified length sequence without overflow; this says nothing about unobserved timing or metadata.

Retained synthetic tool control. The original 24 saved inventory/calendar programs are a separate synthetic demonstration. Discard, shadow and read-only routes preserve the prescribed states by construction; 9/24 live-write runs meet both the narrow goal and protected-bystander condition. This is not evidence for autonomous-agent planning or safe deployment of destructive cover.

S9.1. Retail source, permissions and full statistical results

The pinned τ-bench commit is 59a200c6d575d595120f1cb70fea53cef0632f6b. The instrumented replay preserves the 115 tasks, nine policies, three seeds and two positions (6,210 rows in `retail_units.json`); 97 tasks are eligible. The target is the presence of an attempted reference cancellation. Counts include rejected dispatches. The allowed observer receives ordered tool names and response byte lengths, not argument IDs, response contents or backend state. The tool view is separate from the TLS size observer.

The fixed split takes the first 48 sorted eligible task IDs for fitting (7 positive, 41 negative) and the remaining 49 for testing (11 positive, 38 negative). IDs are recorded in `retail_diagnostics.json`. Every test task contributes all three seeds. For each task we average accuracy across seeds, then resample 49 tasks 10,000 times for accuracy intervals. Balanced-accuracy intervals independently resample within the 11 positive and 38 negative task strata. These intervals condition on this split, fitted models and saved tasks; policy development was not held out. AUROC uses the uncalibrated ridge linear decision score, including ties. Confusion matrices in `retail_observers.json` count 147 task-seed units; they are not 147 independent tasks. Paired $D_2 - D_0$ differences are averaged within task before the task bootstrap.

The known-position filter scores 100% accuracy/BA for each policy and position. The cancellation-count rule scores 100% for blind and other-account cancellation in both positions. Their percentile bootstrap intervals are [100,100] because every observed task is correct; this degeneracy is not a claim about population-level certainty. No identifiers are needed. We do not evaluate an ID-filter attack because IDs are not in the specified view. The legacy model on the other-account after condition recomputes to 71.43%, correcting the previous prose value 68.03%; 68.03% belongs to blind cancellation after insertion.

State metrics and uncertainty. `retail_utility.json` contains task-bootstrap intervals for database equality, full-account equality and equality of reference-changed objects, along with full-census means. The field name `native_reward` is retained only for compatibility with the old script; its calculation is orders/users equality, not an invocation of the full reward/response evaluator. Products are shared read-only by the retail tools. The revised account metric checks all orders owned by the active user, plus the user's record. The reference-changed-object set is constructed from baseline differences, not from all accessed

TABLE S19. LEGACY LEARNED OBSERVER ON THE HELD-OUT RETAIL TASKS. INTERVALS ARE CONDITIONAL TASK-BOOTSTRAP INTERVALS, NOT POLICY-GENERALIZATION GUARANTEES. THE MAJORITY BASELINE IS 77.55% ACCURACY AND 50% BA.

Policy / position	D_0 accuracy [95% CI]	D_2 accuracy [95% CI]	D_2 BA [95% CI]	AUROC
Instruction aware before	51.02 [36.73, 65.31]	73.47 [61.22, 85.71]	47.37 [43.42, 50.00]	0.634
Instruction aware after	51.02 [36.73, 65.31]	71.43 [59.18, 83.67]	52.51 [41.37, 66.15]	0.694
Other-account cancel before	22.45 [12.24, 34.69]	71.43 [59.18, 83.67]	46.05 [40.79, 50.00]	0.497
Other-account cancel after	22.45 [12.24, 34.69]	71.43 [59.18, 83.67]	46.05 [40.79, 50.00]	0.497
Blind cancel before	22.45 [12.24, 34.69]	71.43 [59.18, 83.67]	46.05 [40.79, 50.00]	0.597
Blind cancel after	22.45 [12.24, 34.69]	68.03 [54.42, 80.27]	47.09 [38.16, 58.37]	0.622

objects. Simulator no-cover database equality is a reference-relative 100%, not proof of user-task success.

Error and denominator reconciliation. Item self-write before insertion succeeds and changes state in 173/291 units; all remaining 118 errors are unavailable-variant checks. After insertion, 120 errors are non-pending status and 77 are unavailable variants, leaving 94 successful mutations. The raw error strings are retained. Price-difference checks over the 173 initially successful selections find zero differences, but the API still appends a refund and changes the status. The coincidence 204/291 = 68/97 = 70.10% combines two distinct quantities: preserved reference-changed objects under item self-write and tasks offering an instruction-unmentioned candidate.

For blind cancellation before insertion, database-equal counts by task are: 16:3, 30:3, 31:2, 32:3, 38:3, 54:3, 55:3, 59:2, 66:3, 69:3, 81:2, 99:3. They sum to 33 across 12 tasks. The error-free sensitivity subset is 82 tasks $\times$ 3 seeds = 246 units: 24/246 before and 94/246 after. Simplifying these fractions to 12/123 or 47/123 does not change the underlying sample size. All-unit counts and task IDs are recorded, so repeated seeds are never represented as new independent tasks.

S10. Completed Finite Retrieval Update and Exact Reconciliation

Four Semantic Scholar bulk queries (`S1R`, `S2`, `S3`, `S4`) and two native arXiv searches (`ARXIV-A1`, `ARXIV-A2`) were completed on October 21, 2026, returning 615 occurrences deduplicated to 565 unique records. Combined with the original 60-family October 8 snapshot, the final merged index contains 82 families: 58 core (20 workflow-origin, 38 inherited; 53 non-diagnostic and 5 Type IV diagnostic) and 24 adjacent.

S10.1. Executable queries and pre/post-audit disposition accounting

The Semantic Scholar bulk queries (2023–2026) and native arXiv CS searches (January 1, 2023–October 21, 2026) are:

S1R (116): (agent | workflow) + (LLM | "language model") + ("side channel" | "side-channel" | fingerprinting).
S2 (120): (LLM | "language model") + (traffic | streaming) + (leakage | side-channel | fingerprinting).
S3 (15): (LLM | "language model") + (padding | "re-emission") + (privacy | leakage | side-channel).
S4 (287): (agent | LLM) + (decoy | "cover traffic" | "fake actions" | "cover actions").
ARXIV-A1 (50): `"side-channel" AND "large language"`; ARXIV-A2 (27): `"side-channel" AND agent`.

Table S21 provides the complete item-by-item reconciliation before and after the 204-record exclusion audit (177 high-risk pattern matches plus 27 reason-stratified samples, seed `20261021`). Before the exclusion audit, screening identified 221 C2/C3 exclusions (`EXCLUDE_NO_HIDDEN_AUXILIARY_TARGET`), 14 new core families, 23 existing core families, and 28 source-reviewed adjacent contexts. The exclusion audit rerouted exactly four records out of the 221 C2/C3 exclusions ($221 - 4 = 217$):

1) `SUP-0192` (MIRAGE [1]): rerouted to `NEW_CORE_SOURCE_VERIFIED` ($14 + 1 = 15$ final new core families, all 15 incorporated into the 82-family index via `addition.csv`).
2) `SUP-0552` (InputSnatch arXiv version [35]): rerouted to `EXISTING_CORE` ($23 + 1 = 24$ existing core recoveries; merged as an alias of the already-indexed snapshot family `inputsnatch2024`, adding $+0$ net new families).
3) `SUP-0191` (LLM Self-Recognition) and `SUP-0424` (Reverse-Engineering Model Editing): rerouted to `ADJACENT_SOURCE_REVIEWED` ($28 + 2 = 30$ final source-reviewed adjacent contexts). Of these 30 source-reviewed adjacent contexts, 7 boundary-defining studies (`addition-adjacent.csv`) are incorporated into the 82-family index ($17 + 7 = 24$ adjacent), while the other 23 (including `SUP-0191` and `SUP-0424`) remain in the 565-record disposition ledger as non-indexed context.

Thus the family transition is strictly additive and self-consistent: 60 (snapshot) + 15 (14 pre-audit + 1 audited new core) + 7 (selected adjacent) = 82 families ($43 + 15 = 58$ core; $17 + 7 = 24$ adjacent).

TABLE S20. REFERENCE-RELATIVE STATE EQUALITY ON 97 ELIGIBLE RETAIL TASKS. ENTRIES ARE PERCENTAGES WITH CONDITIONAL 95% TASK-BOOTSTRAP INTERVALS. NO-COVER, DISCARD, SHADOW, READ-ONLY AND ADDRESS-WRITE CONDITIONS ARE 100 [100,100] FOR ALL THREE SCOPES.

Policy / position	Database equality	Full-account equality	Reference-changed-object equality
Item self-write before	40.55 [31.62, 50.17]	40.55 [31.62, 50.17]	70.10 [61.51, 78.35]
Item self-write after	67.70 [59.10, 76.29]	67.70 [59.10, 76.29]	97.25 [93.81, 99.66]
Instruction aware before	35.05 [25.77, 44.33]	35.05 [25.77, 44.33]	78.35 [70.10, 86.60]
Instruction aware after	46.39 [36.08, 56.70]	46.39 [36.08, 56.70]	89.69 [83.51, 94.85]
Other-account cancellation before	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	100.00 [100.00, 100.00]
Other-account cancellation after	0.00 [0.00, 0.00]	100.00 [100.00, 100.00]	100.00 [100.00, 100.00]
Blind cancellation before	11.34 [5.84, 17.87]	11.34 [5.84, 17.87]	62.54 [53.61, 71.48]
Blind cancellation after	41.24 [32.30, 50.17]	41.24 [32.30, 50.17]	92.44 [87.63, 96.56]

TABLE S21. POST-AUDIT DISPOSITION FLOW FOR THE COMPLETED 565-RECORD SUPPLEMENTARY FRAME (SEE TABLE S3 FOR PRE-AUDIT VS. POST-AUDIT Δ).

Stage or final post-audit route	Count
Complete-query occurrences / duplicates merged	615 / 50
Unique records with an explicit route	565
Excluded: no hidden auxiliary target ($221 - 4$ rerouted)	217
Excluded: outside LLM execution / different target	156 / 19
Existing core ($23 + 1$ alias) / adjacent recovered	24 / 1
New source-bound core families ($14 + 1$ MIRAGE; all indexed)	15
Adjacent source-reviewed ($28 + 2$; 7 indexed) / metadata-only	30 / 28
Other security background / reviewed background / proposal	53 / 4 / 1
Missing abstract / unavailable primary full text	16 / 1

S10.2. Seed overlap clarification and dual-coding reliability

As noted in Main Paper Section 2, the 24 fixed known papers (`known24`) are a 24-paper core subset of our 40 initial citation-tracing seeds (`seed_n=24` in `check_known24_offline.py`), not an external holdout. Their 17/24 query recovery (and 25/60 snapshot recovery) measures lexical overlap between the six Boolean keyword queries and our seed-and-snowball corpus. Supplied second-coder annotations of a 100-record stratified subsample of the 565-record frame (`m7_dual_coding_screening.csv`) achieved 93.0% three-class agreement ($\kappa = 0.886$, 95% CI $[0.801, 0.963]$) and 95.0% binary Core-inclusion agreement ($\kappa = 0.880$, CI $[0.764, 0.975]$).

S11. Retained buffered-TLS and fixed-position retail studies

This section preserves, unchanged in substance, the first-revision main-text account of the two controlled studies that Section 8 of the main paper now summarizes. Section S12 reports the second-round tests E5–E7.

The two studies test distinct observers. The transport observer sees TLS application-record lengths on a selected link. The tool-dispatch observer sees ordered tool names, write counts and response byte lengths, including failed calls; it is an instrumented dispatch view, not an encrypted-network observer. Neither experiment combines these views or runs an autonomous agent. Saved outputs and reference programs let us vary the specific boundary while holding the prescribed workload fixed.

S11.1. Transport placement and workload-matched framing

For 24 public topics, the saved corpus contains one prompted brief and one prompted expanded `gpt-6-luna` response (48 outputs; 12 training and 12 disjoint test topics). These are designed prompt conditions, not a sample of natural task lengths. The original placement experiment passes text through two TLS 1.2 connections: the frontend decodes and removes upstream framing before sending new terminal chunks. Exact plaintext hashes are checked. Equalizing the upstream length projection leaves the unpadded terminal projection separable (100% verbosity classification); padding both projections reduces the evaluated classifier to 50%. This reproduces the receiving-boundary mechanism identified by NetEcho rather than discovering a new failure mode.

The revised framing replay uses `ECDHE-RSA-AES128-GCM-SHA256` sockets and parses application-data records. T1 contrasts brief versus full expanded responses. T2 contrasts the brief with the first two period-delimited sentences of the expanded response. T3 contrasts the first two versus first three sentences of the same expanded response. Thus T2/T3 are derived prefixes, with two examples per topic, not independently generated explanations. T4 relabels both brief and expanded responses by topic domain; it has 24 test outputs from 12 topics, not 12 expanded outputs. Its unpadded accuracy is 45.8% (95% topic-bootstrap interval 25.0–66.7%); its test majority-class baseline is 58.3%. T4 is a weak-signal control, not a defense success.

Every cost ratio divides defended application-record bytes by unpadded application-record bytes for the *same texts*, over all 48 examples in that workload. The count includes TLS record headers and encryption overhead; handshakes, TCP/IP, upstream bytes and latency are excluded. The 4-byte length prefix is included. Buckets of 256/512 bytes use 128-byte writes; 1024-byte buckets use 256-byte

TABLE S22. CORRECTED TLS APPLICATION-RECORD RESULTS. COSTS USE EACH WORKLOAD'S OWN UNPROTECTED TEXTS (ALL 48 OUTPUTS); ACCURACY USES 24 TEST OUTPUTS IN 12 TOPICS. INTERVALS RESAMPLE TOPICS.

A. Two-link placement (T1; retained experiment)

Protection	Upstream acc.	Terminal acc.	Terminal bytes
None	100.0	100.0	720.96
Upstream only	50.0	100.0	720.96
Both, loose cap	50.0	50.0	16,848

B. Matched workload costs and defended-fit accuracy

Workload / framing	Mean bytes	Ratio	Acc. (%)
T1: unprotected	720.96	1.00	100.0
T1: 256B bucket	856.96	1.19	100.0
T1: 512B bucket	1,085.92	1.51	100.0
T1: 1024B bucket	1,401.25	1.94	70.8
T1: training-fitted cap	1,570.00	2.18	50.0
T1: loose cap	16,848.00	23.37	50.0
T2: unprotected	272.48	1.00	100.0
T2: 256B bucket	438.29	1.61	87.5
T2: 512B bucket	628.00	2.30	50.0
T2: 1024B bucket	1,140.00	4.18	50.0
T3: unprotected	456.33	1.00	91.7
T3: 256B bucket	608.38	1.33	62.5
T3: 512B bucket	654.17	1.43	50.0
T3: 1024B bucket	1,140.00	2.50	50.0
T4: unprotected	720.96	1.00	45.8

C. Adaptation on identical defended T1 tests

Defense	D_0 (%)	D_2 (%)	Gain [95% CI], pp
+512B offset	50.0	100.0	50.0 [50.0, 50.0]
1024B bucket	50.0	70.8	20.8 [8.3, 33.3]
$U(0, 512)$B	86.4	99.9	13.5 [10.7, 16.2]
$U(0, 1024)$B	68.2	89.2	21.1 [17.2, 24.8]
Loose cap	50.0	50.0	0.0 [0.0, 0.0]
Training-fitted cap	50.0	50.0	0.0 [0.0, 0.0]

Panel C uses 12 training pairs. Four-pair accuracy is the same except under $U(0, 1024)$B, where it is 90.2%. Random padding averages 50 draws within each topic; these are not 50 independent corpora. T2/T3 are sentence-prefix constructions. T4 relabels both original response lengths and has a 58.3% test majority baseline. The cap equalizes only the specified record-length projection when no message overflows.

writes. Consequently the 1024-byte single-quantum wire size is 1,140 bytes, not 1,082. The workload-matched 512-byte ratios for T2/T3 are 2.30×/1.43×; dividing both by T1's 720.96-byte denominator would yield the invalid earlier ratios below one.

A tighter comparison selects a constant application-frame cap using *training lengths only*: round the largest framed training response (1,194 bytes) up to a multiple of 128, yielding 1,280 bytes. Ten 128-byte records cost 1,570 wire bytes (2.18× T1). Longer messages would emit additional 128-byte records without truncation; no test message overflows. This produces identical length sequences on the saved T1 test set and 50% accuracy for the evaluated classifiers. It is a feasible cap for these data, not a globally optimal padding protocol or a guarantee for future lengths. The retained 16,848-byte cap (23.37×) is only a deliberately loose control.

S11.2. Matched attacker adaptation

We use D_0 for an attacker fitted on unprotected training data and D_2 for one fitted on defended training data; both are evaluated on the identical defended test examples. The threshold uses total TLS record payload bytes. D_2 uses the first four or all twelve training topic pairs; these budgets do not represent multiple independently sampled training sets. At a 1024-byte bucket, D_0 constantly predicts the expanded class and scores 50%, while D_2 scores 70.8% (58.3–83.3%). An offset of 512 bytes permits 100% adapted accuracy. This demonstrates why a shifted D_0 alone is insufficient, without claiming novelty for adaptive evaluation.

Random-tail padding is evaluated over 50 draws per output. The first draw is socket measured and the remaining 49 use the record-length accounting checked against the socket replay. Each topic's paired correctness is averaged over its outputs and draws before 10,000 topic-bootstrap resamples (seed 20261021). Fractional accuracies therefore do not imply additional independent topics; the intervals condition on these saved outputs and fitted attackers. Positive adaptation gains span 13.5–50.0 percentage points; both constant caps have zero gain.

No measured accuracy of 50% proves absence of all leakage. The constant-cap construction makes only the specified record-length sequence constant when there is no overflow. Timing, connection behavior, request metadata and content are outside that projection. No sequence/timing attack, TLS 1.3 native padding, QUIC or live token/SSE stream is evaluated.

S11.3. Retail execution: state scope and a visible-call counterexample

We replay the 115 public reference programs in the pinned τ-bench retail simulator. There is no LLM planner or user simulator in this experiment. Ninety-seven tasks have an initially pending order owned by the task user; 18 have none and receive no cover. Three seeds select orders, and cover is inserted before or after the prescribed program. Utility uses 97 task clusters, each with three repeated units per position. Policies include no cover, discard, shadow execution, read only, address self-write, item self-write, instruction-aware selection, other-account cancellation and blind cancellation. Destructive conditions are local negative controls, not proposed deployment policies.

Separate the outcome scopes. Table S23 reports reference database equality, full active-account equality and equality on objects *changed* by the reference program. The last scope does not include every read/accessed object. The database criterion corresponds to the state part of the native evaluator; the extension script does not evaluate generated response correctness and we do not relabel its state equality as an end-to-end native task reward. All-account equality now checks every order owned by the active user and that user's record, rather than only initially pending orders.

Address self-writes preserve the reference state in this simulator. Item self-writes mutate state in 173/291 pre-

TABLE S23. RETAIL STATE AND OBSERVER RESULTS. STATE COLUMNS USE 97 ELIGIBLE TASKS (291 UNITS PER POSITION); OBSERVER COLUMNS USE 49 HELD-OUT TASKS (147 UNITS). PERCENTAGES ARE TASK MEANS ACROSS THREE SEEDS. B/A DENOTES BEFORE/AFTER.

A. Distinct state scopes (reference execution as comparator)

Policy / position	DB eq.	Account eq.	Object eq.	Mutated
No cover B/A	100.00	100.00	100.00	0.00
Discard B/A	100.00	100.00	100.00	0.00
Shadow B/A	100.00	100.00	100.00	0.00
Read only B/A	100.00	100.00	100.00	0.00
Address self-write B/A	100.00	100.00	100.00	0.00
Item self-write B	40.55	40.55	70.10	59.45
Item self-write A	67.70	67.70	97.25	32.30
Instruction aware B	35.05	35.05	78.35	70.10
Instruction aware A	46.39	46.39	89.69	53.61
Other-account cancel B	0.00	100.00	100.00	100.00
Other-account cancel A	0.00	100.00	100.00	100.00
Blind cancel B	11.34	11.34	62.54	100.00
Blind cancel A	41.24	41.24	92.44	58.76

B. Legacy learned observer versus allowed-view rules

Policy / position	D_0 acc.	D_2 acc.	D_2 BA	Rule acc.
Majority (all conditions)	77.55	77.55	50.00	—
Instruction aware B	51.02	73.47	47.37	100.00
Instruction aware A	51.02	71.43	52.51	100.00
Other-account cancel B	22.45	71.43	46.05	100.00
Other-account cancel A	22.45	71.43	46.05	100.00
Blind cancel B	22.45	71.43	46.05	100.00
Blind cancel A	22.45	68.03	47.09	100.00

DB eq.: equality of orders/users against reference (products are read-only), the native reward's database criterion, not an output-response/authorization check. Account eq.: all orders owned by the active user plus that user's record. Object eq.: objects changed by the reference program, not all accessed objects. Mutated: cover itself changes state. Other-account calls yield 0% DB equality but 100% active-account equality. BA: balanced accuracy. Rule: remove the known first/last cover call, then inspect cancellation verbs; blind/other-account cancellation also admits the $N_{\text{cancel}} \geq 2$ rule. Both rules attain 100% acc./BA in their stated conditions. Other non-cancellation policies retain 100% verb inference. Full intervals, AUROC and confusion matrices appear in Supplement S9 and the JSON artifact.

insertion units (59.45%): the tool changes the status to `pending (item modified)` and appends a refund entry even when the item identifier and price are unchanged. The remaining 118/291 calls fail because the requested variant is unavailable, explaining why this mutation rate is below blind cancellation's 291/291 successful calls. This is a state-preservation counterexample from one API, not a prevalence estimate or a discovery about idempotence. Formally, a no-change requirement $f(s, a) = s$ differs from repeated-call idempotence $f(f(s, a), a) = f(s, a)$.

Instruction-aware cover uses only instruction text and the initial order table for selection, with address self-write as fallback. It preserves reference-changed objects in 78.35%/89.69% of before/after units, but preserves the full account and database in only 35.05%/46.39%. Cancelling an unmentioned order can therefore damage a bystander resource. The heuristic was developed using the available task collection; it has no held-out policy-development evaluation. Its 68/97 availability (70.10%) and the item-self-write object equality of 204/291 (70.10%) are separate counts that happen to yield the same percentage.

The other-account condition selects a pending order belonging to another simulated user. It is not a provisioned dedicated decoy account. It preserves the active user's complete account and reference-changed objects in these tasks, while changing the shared database in every eligible unit (0% database equality). Thus account isolation of an effect cannot be described as whole-database preservation or an authorized production defense.

Observer permissions and simple attacks. The secret label is whether the reference program *attempts* a cancellation, including a call that may fail after cover execution. The observer sees the ordered dispatched tool names, write counts and response lengths; failed calls remain visible. It sees no arguments, user/order identifiers or database state. We do not infer that a passive TLS observer has these permissions. A backend with plaintext state would instead need a different confidentiality target.

The fixed task split contains 48 training tasks (7 positive, 41 negative) and 49 test tasks (11 positive, 38 negative); seeds stay within tasks. Always predicting no cancellation gives 77.55% test accuracy and 50% balanced accuracy. The legacy D_2 linear classifier uses cancellation presence, write count and write-response bytes. Its 68.03–73.47% accuracy under cancellation cover is below the majority baseline, and its balanced accuracy ranges from 46.05% to 52.51%. An increase over D_0 is therefore insufficient evidence for its claimed privacy interpretation.

For blind and other-account cancellation, exactly one extra cancellation is dispatched in every eligible session. The permitted rule $\widehat{y} = \mathbf{1}[N_{\text{cancel}} \geq 2]$ recovers the reference label at 100% accuracy and balanced accuracy on all 49 test tasks and three seeds, before and after insertion. For any policy with one live cover call at a known first/last position, removing that call and testing for a remaining cancellation also gives 100%. It needs no identifiers, no response content and no training labels. These are measured finite-sample results and a structural property of the fixed insertion design, not proof of general attack performance. Identifier filtering is unnecessary in this view and is not claimed as an evaluated attack.

Terminal equality does not establish path properties. Blind pre-insertion cancellation leaves the final database equal to reference in 33/291 units across 12 tasks, even though it preempts the later reference cancellation. The exact per-task counts are in Supplement S9. After insertion, 120/291 cover calls are rejected because reference execution already changed the order's status, and those rejections explain the 41.24% equality rate. Rejection is not successful benign cover. In the 82 tasks with no reference execution error (246 units), before/after equality is 24/246 (9.76%) and 94/246 (38.21%). These results illustrate the limits of endpoint scoring; consent and authorization were not measured.

S12. Second-round transfer tests (E5–E7)

E5 and individual-attacker E6 results below retain the R2 measurements. Current E6/E7 suite summaries use post-review reanalysis in `experiments/sp2027-r3/`; historical registration and outputs remain in `experiments/sp2027-r2/`. A suite maximum is recomputed in each of 2,000 common task-bootstrap draws. Intervals condition on fixed out-of-fold predictions and omit retraining/search uncertainty. Meets: upper BA bound < 0.60 and maximum point AUROC < 0.65; Above: lower BA bound > 0.60; otherwise inconclusive. This is a finite-suite criterion, not a security guarantee.

TABLE S24. E5 PER ATTACKER WITH 95% TOPIC-BOOTSTRAP INTERVALS (12 TEST TOPICS). WIRE: MEAN APPLICATION-RECORD BYTES INCL. 5-BYTE HEADERS. AMB.: MEAN SIZE OF THE NEAREST-SEQUENCE TIE SET.

TLS	Target	Framing	Wire	D_0 total	D_2 total	D_2 count	D_2 sequence	ID	Amb.
1.2	T1	Buffered, unpadded	721	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	87.5	1.3
1.2	T1	Buffered, training cap	1570	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	2.1	48.0
1.2	T1	Stream, per token	15038	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0	1.0
1.2	T1	+ random event pad	19194	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	62.3	1.0
1.2	T1	+ constant event size	18255	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	58.3	2.3
1.2	T1	8-token batches, const.	3040	50.0 [50.0,50.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	20.8	9.4
1.2	T1	Const. events + count cap	36092	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	6.3	44.1
1.2	T3	Buffered, unpadded	456	91.7 [79.2,100.0]	91.7 [79.2,100.0]	91.7 [79.2,100.0]	91.7 [79.2,100.0]	91.7	1.2
1.2	T3	Buffered, training cap	635	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	4.2	44.2
1.2	T3	Stream, per token	9713	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	100.0	1.0
1.2	T3	+ random event pad	12398	71.2 [58.3,83.8]	87.9 [75.8,99.2]	87.5 [75.0,100.0]	87.9 [76.2,98.3]	72.5	1.0
1.2	T3	+ constant event size	11803	79.2 [66.7,91.7]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	70.8	1.8
1.2	T3	8-token batches, const.	2023	50.0 [50.0,50.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	18.8	8.4
1.2	T3	Const. events + count cap	15411	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	12.5	38.6
1.3	T1	Buffered, unpadded	686	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	87.5	1.3
1.3	T1	Buffered, training cap	1500	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	2.1	48.0
1.3	T1	Stream, per token	14294	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0	1.0
1.3	T1	+ random event pad	18450	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	62.3	1.0
1.3	T1	+ constant event size	17511	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	58.3	2.3
1.3	T1	8-token batches, const.	2938	50.0 [50.0,50.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	20.8	9.4
1.3	T1	Const. events + count cap	34626	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	6.3	44.1
1.3	T3	Buffered, unpadded	433	91.7 [79.2,100.0]	91.7 [79.2,100.0]	91.7 [79.2,100.0]	83.3 [70.8,95.8]	91.7	1.2
1.3	T3	Buffered, training cap	606	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	4.2	44.2
1.3	T3	Stream, per token	9230	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	100.0	1.0
1.3	T3	+ random event pad	11915	68.8 [56.7,81.7]	87.9 [75.8,99.2]	87.5 [75.0,100.0]	87.5 [75.4,98.3]	72.5	1.0
1.3	T3	+ constant event size	11321	79.2 [66.7,91.7]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	70.8	1.8
1.3	T3	8-token batches, const.	1953	50.0 [50.0,50.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	18.8	8.4
1.3	T3	Const. events + count cap	14782	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	12.5	38.6

TABLE S25. E6 DISPATCH VIEW: BALANCED ACCURACY (%) WITH TASK-CLUSTER BOOTSTRAP INTERVALS (115 TASKS, 3 SEEDS PER TASK), BEST AUROC, ACTIVE-ACCOUNT EQUALITY AND COVER-CALL ERROR RATE.

Target	Cover	D_0 GB	D_2 LR	D_2 GB	Count rule	AUROC	Acct. eq.	Cover err.
cancel	No cover	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
cancel	Cancel, p=0.5	74.6 [71.6,77.4]	84.5 [75.9,92.1]	86.1 [78.4,93.9]	83.3 [75.0,91.2]	0.95	100.0	0.0
cancel	Cancel, p=1	50.0 [50.0,50.0]	97.1 [92.4,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
cancel	Cancel, p=1, live execution	50.0 [50.0,50.0]	98.1 [95.4,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
cancel	Cancel count $\rightarrow$ 2	50.0 [50.0,50.0]	76.9 [64.8,88.0]	69.4 [57.6,81.7]	50.0 [50.0,50.0]	0.85	100.0	2.8
cancel	One decoy program	91.9 [89.8,94.2]	68.1 [60.2,76.7]	66.0 [60.1,72.2]	91.9 [89.8,94.2]	0.92	100.0	3.0
cancel	Two decoy programs	83.5 [80.5,86.5]	55.9 [48.4,63.6]	57.0 [49.9,64.1]	83.5 [80.5,86.5]	0.86	100.0	4.3
cancel	Shape equalization	50.0 [50.0,50.0]	44.8 [37.6,52.4]	49.2 [44.0,54.8]	50.0 [50.0,50.0]	0.50	100.0	5.7
return	No cover	100.0 [100.0,100.0]	97.8 [94.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
return	Cancel, p=0.5	100.0 [100.0,100.0]	98.0 [94.3,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
return	Cancel, p=1	100.0 [100.0,100.0]	97.5 [94.8,99.6]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
return	Cancel, p=1, live execution	100.0 [100.0,100.0]	98.0 [94.3,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
return	Cancel count $\rightarrow$ 2	100.0 [100.0,100.0]	98.0 [95.4,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	2.8
return	One decoy program	85.9 [82.9,89.0]	73.2 [66.6,79.5]	71.1 [65.5,76.8]	85.9 [82.9,89.0]	0.88	100.0	3.0
return	Two decoy programs	78.1 [75.0,81.2]	70.6 [64.1,77.3]	69.1 [62.9,75.0]	78.1 [75.0,81.2]	0.85	100.0	4.3
return	Shape equalization	50.0 [50.0,50.0]	54.5 [48.4,60.3]	50.3 [45.4,55.5]	50.0 [50.0,50.0]	0.53	100.0	5.7

TABLE S26. CORRECTED E6 SUITE MAXIMA AND TASK-BOOTSTRAP INTERVALS. INDIVIDUAL-ATTACKER RESULTS ABOVE ARE UNCHANGED. AUROC IS ALSO MAXIMIZED OVER THE EVALUATED SUITE.

Target	Cover	Max BA [95% CI]	Max AUROC	Criterion
cancel	No cover	100.0 [100.0,100.0]	1.00	Above
cancel	Cancel, p=0.5	86.1 [80.9,94.2]	0.95	Above
cancel	Cancel, p=1	100.0 [100.0,100.0]	1.00	Above
cancel	Cancel, p=1, live	100.0 [100.0,100.0]	1.00	Above
cancel	Cancel count $\rightarrow$ 2	76.9 [67.0,88.1]	0.85	Above
cancel	One decoy	91.9 [89.8,94.2]	0.92	Above
cancel	Two decoys	83.5 [80.5,86.5]	0.86	Above
cancel	Shape	50.0 [50.0,54.8]	0.50	Meets
return	No cover	100.0 [100.0,100.0]	1.00	Above
return	Cancel, p=0.5	100.0 [100.0,100.0]	1.00	Above
return	Cancel, p=1	100.0 [100.0,100.0]	1.00	Above
return	Cancel, p=1, live	100.0 [100.0,100.0]	1.00	Above
return	Cancel count $\rightarrow$ 2	100.0 [100.0,100.0]	1.00	Above
return	One decoy	85.9 [82.9,89.0]	0.88	Above
return	Two decoys	78.1 [75.0,81.2]	0.85	Above
return	Shape	54.5 [50.0,60.4]	0.53	Inconcl.

TABLE S27. CORRECTED E7 SUITE MAXIMA (4,140 SAVED TLS SESSIONS). DECODER: HELD-OUT REQUEST/RESPONSE SIZE-TO-TOOL ACCURACY; IT DIFFERS FROM REQUEST-ONLY IDENTIFICATION. COSTS USE RAW/NO-COVER AS DENOMINATOR. DEFENSE CALIBRATION USES THE FULL PUBLIC TASK POPULATION.

Target	Framing	Cover	Max BA [95% CI]	AUROC	Criterion	Decoder	Bytes×	Calls×
cancel	raw	No cover	100.0 [100.0,100.0]	1.00	Above	92.8	1.00	1.00
cancel	raw	Cancel count $\rightarrow$ 2	72.7 [63.4,84.5]	0.81	Above	94.7	1.38	1.35
cancel	raw	One decoy	91.9 [89.8,94.2]	0.92	Above	99.7	2.03	2.01
cancel	raw	Shape	51.2 [50.0,58.8]	0.54	Meets	100.0	7.56	7.71
cancel	bucket1024	No cover	60.1 [50.0,73.3]	0.67	Inconcl.	35.2	2.42	1.00
cancel	bucket1024	Cancel count $\rightarrow$ 2	67.5 [58.5,78.9]	0.70	Inconcl.	33.5	3.25	1.35
cancel	bucket1024	One decoy	60.0 [53.4,68.7]	0.59	Inconcl.	35.5	4.90	2.01
cancel	bucket1024	Shape	55.3 [52.2,63.6]	0.59	Inconcl.	26.5	18.56	7.71
cancel	const	No cover	59.7 [50.0,72.3]	0.62	Inconcl.	30.4	3.41	1.00
cancel	const	Cancel count $\rightarrow$ 2	64.9 [54.9,76.7]	0.66	Inconcl.	30.0	4.61	1.35
cancel	const	One decoy	59.7 [52.6,67.4]	0.59	Inconcl.	30.5	6.87	2.01
cancel	const	Shape	54.4 [50.0,62.5]	0.50	Inconcl.	21.6	26.32	7.71
return	raw	No cover	95.7 [90.7,99.4]	0.96	Above	93.8	1.00	1.00
return	raw	Cancel count $\rightarrow$ 2	95.7 [90.6,99.4]	0.96	Above	95.4	1.38	1.35
return	raw	One decoy	85.9 [82.9,89.0]	0.88	Above	100.0	2.03	2.01
return	raw	Shape	60.5 [54.9,66.4]	0.61	Inconcl.	100.0	7.56	7.71
return	bucket1024	No cover	73.9 [66.6,83.1]	0.76	Above	35.2	2.42	1.00
return	bucket1024	Cancel count $\rightarrow$ 2	66.0 [59.8,75.8]	0.70	Inconcl.	33.5	3.25	1.35
return	bucket1024	One decoy	59.1 [54.1,66.8]	0.60	Inconcl.	35.5	4.90	2.01
return	bucket1024	Shape	50.6 [50.0,57.3]	0.52	Meets	26.5	18.56	7.71
return	const	No cover	53.6 [50.0,63.9]	0.55	Inconcl.	30.4	3.41	1.00
return	const	Cancel count $\rightarrow$ 2	59.1 [52.3,68.6]	0.61	Inconcl.	30.0	4.61	1.35
return	const	One decoy	56.8 [51.4,63.6]	0.57	Inconcl.	30.5	6.87	2.01
return	const	Shape	50.0 [50.0,55.1]	0.50	Meets	21.6	26.32	7.71

TABLE S28. REQUEST-ONLY TOOL IDENTIFICATION: TRAINING-FOLD SIZE TABLE, UNSEEN-SIZE FALLBACK TO THE TRAINING MAJORITY. EACH OF 582 REFERENCE CALLS IS TESTED ONCE. THE 95.7% RAW IN-SAMPLE PURITY IS EXCLUDED FROM HELD-OUT ACCURACY.

Task stratification	Framing	Held-out accuracy (%)	Training-majority baseline (%)
cancel	raw	89.9	29.4
cancel	bucket1024	29.4	29.4
cancel	const	29.4	29.4
return	raw	91.4	29.4
return	bucket1024	29.4	29.4
return	const	29.4	29.4

Training-only tool-count maxima overflow on five distinct held-out tasks (22, 28, 40, 67, 87) under either target split. This is a feasibility audit, not a rerun of defense efficacy. Full fold membership and overflow records are released. No tested network configuration meets the criterion for both targets.

TABLE S29. CURRENT REASSESSMENT OF THE SEVEN AUTHOR-WRITTEN JUDGMENTS RECORDED BEFORE E5–E7. HISTORICAL OUTCOME LABELS ARE PRESERVED IN THE ARTIFACT, NOT COUNTED AS AUDIT ACCURACY.

ID	Claim under examination	Current evidence and qualification
P1	Buffered cost transfers to a stream	Emitter mismatch supported: per-event padding leaves T1 at 100%; constant-shape stream costs 2.42×.
P2	Low D_0 establishes protection	Counterexample supported: batching and cancellation cover give D_0 50%, D_2 100%.
P3	Per-event padding prevents identification	Counterexample supported: top-1 58–62% versus 2.1% chance in the closed candidate set.
P4	Equal cancellation count hides cancellation	Residual leakage supported: count rule 50%; maximum BA 76.9%.
P5	Dispatch protection transfers to the tool link	Protected-source premise unmet: count/decoy policies already leak at dispatch. Network BA 72.7–95.7% shows persistence.
P6	Framing alone protects intent	Bucketed return remains above threshold; remaining framing-only cells inconclusive.
P7	Joint cover/framing protects both targets	Not established: no configuration meets both target criteria. Joint constant framing: 50.0–54.4% BA; 26.3× bytes, 7.7× calls.

S13. Stand-in planner pipeline (E8) and online cover

`experiments/sp2027-r2/agent/` runs tool-calling agents and a user simulator on the 115 retail tasks, applies *online* cover (decided before each real call from the history only, with a final flush) and replays every session over TLS. Because sandboxed cover and framing never reach the agent, one trajectory per task and seed serves all policies. No LLM was available for this revision; the rows below use stand-in planners and exercise the pipeline. The scripted planner replays reference programs (native reward 1.00); the noisy planner inserts extra reads before each reference call with probability 0.3 (7.2 versus 5.1 real calls per session). These are synthetic results, not agent measurements.

TABLE S30. E8 DIAGNOSTICS REPORT POINT ESTIMATES ONLY; NO SUITE-MAXIMUM INTERVAL WAS RECOMPUTED FOR ONLINE RUNS. CURRENT OFFLINE E6/E7 INTERVALS ARE IN S12. BEST HELD-OUT BALANCED ACCURACY (%) POINT ESTIMATES FOR OFFLINE VERSUS ONLINE COVER. CONST.: CONSTANT FRAMING ON THE AGENT–TOOL LINK. CALLS+: COVER CALLS PER REAL CALL. OVERFLOW: SESSIONS WHOSE REAL CALLS EXCEED THE SHAPE TEMPLATE OR THE CANCELLATION TARGET.

Planner / cover timing	Cover	Dispatch C	Dispatch R	Const. C	Const. R	Calls+	Overflow (%)
Offline (E6/E7)	cancel const2	76.9	100.0	64.9	59.1	0.35	0.0
Offline (E6/E7)	decoy1	91.9	85.9	59.7	56.8	1.01	0.0
Offline (E6/E7)	shape eq	50.0	54.5	54.4	50.0	6.71	0.0
Online, scripted	cancel const2	79.6	100.0	63.0	59.1	0.36	3.8
Online, scripted	decoy1	92.3	88.6	51.2	56.3	0.98	0.0
Online, scripted	shape eq	63.8	55.1	57.5	55.9	6.73	0.0
Online, noisy scripted	cancel const2	79.1	100.0	66.0	60.0	0.25	4.1
Online, noisy scripted	decoy1	92.3	88.6	51.8	56.2	0.69	0.0
Online, noisy scripted	shape eq	55.9	53.9	60.5	56.2	4.52	20.9

The preceding E8 rows and availability statement describe the historical stand-in study. S14 reports the subsequent real-model collection and training-only calibration.

S14. Complete Real-Agent Evaluation with Training-Only Calibration (E9)

S14.1. Cohort and execution

All 690 episodes in the frozen protocol are complete: 115 tasks, two prompt arms and three repetition identifiers. The fixed split assigns 69 tasks from 29 customers to training (414 episodes) and 46 tasks from 24 disjoint customers to testing (276 episodes). Every test task has all six prompt/repetition cells, and all failures and handoffs remain in the analysis. Training finished before defense calibration was frozen and test inference began. This section updates the earlier 663-episode publication snapshot to the full cohort; the immutable snapshot and the previous manuscript remain available in the archived release. No model configuration, prompt, split or defense parameter changed for this completion.

GPT-6.1 Sol with low reasoning effort supplies the planner and a separate user simulator through Codex app-server. The planner receives the retail policy, public tool schemas, interaction history and actual sandbox observations, and emits one structured action at each step. Reference actions and reward targets are used only for scoring. Each decision has a fresh ephemeral context without host tools or a host environment. The harness executes decisions in the pinned public retail simulator, with a 30-step limit. Repetition identifiers are cache namespaces; provider random seeds, temperature and an immutable weight snapshot are unavailable.

S14.2. Prompts and defense calibration

The baseline prompt is the retail policy. The rules prompt adds one customer-details read after identification and before order access, an order read before each order write, and avoidance of unnecessary repeated reads. It preserves authorization and task requirements and prohibits unrelated operations for hiding intent. Both arms are evaluated under raw transport, padding only, online shape cover only, and joint shape cover with constant framing.

Only training trajectories from both prompts determine shared per-tool count maxima, donor pools and size caps. Donor calls exclude the active customer. Request caps are the largest training request plus four bytes rounded to a 256-byte multiple; response caps use the training 95th percentile plus four bytes rounded to a 1024-byte multiple, including training donor responses. Frozen caps are 1,024 and 3,072 bytes. The online dispatcher uses only past real-call counts, inserts a shadow call with probability 0.6 when a deficit exists and flushes remaining deficits at episode end. Overflow calls pass through; oversize messages expand to a 1024-byte bucket. Test data never recalibrate these parameters.

Shadow calls execute on isolated cloned state; their observations are suppressed and real calls/responses pass through unchanged. For deterministic tools with no timing-dependent side effects, projecting away cover gives the same planner-visible history under all program conditions. The replay therefore isolates transport intervention on the same realized actions, with reward preserved by the isolation invariant. All 690 real-call projections agree across the 2,760 saved sessions. Online means scheduling from past calls; interactive wall-clock delay and shared-resource feedback remain outside this replay measurement. A byte-transparent relay captures actual serialized requests and sandbox responses over local TLS 1.2.

S14.3. Attackers and uncertainty

The frozen suite includes a training-majority baseline, clean-trained gradient boosting, defense-trained logistic regression and gradient boosting, a dispatch count rule, and network size-decoding learners and rules. All fitted parameters, thresholds and decoder tables use training data only. Targets are cancellation and return intent from the reference task; native task reward is scored separately using final database equality and required response strings. Points weight tasks equally. Intervals use 2,000 common customer-cluster bootstrap draws, retaining each customer's tasks and repetitions together and recomputing the suite maximum in each draw. Intervals condition on fitted attackers. Prompt contrasts pair task/repetition identifiers, without assuming identical conversations or shared model randomness.

S14.4. Utility, overhead and prompt contrasts

Cancellation has 6 positive tasks from 5 customers; return has 11 positive tasks from 7 customers. Single-class draws excluded from 2,000 replicates: 6 for cancellation and 0 for return.

Baseline native reward is 110/138 (79.7%); rules reward is 105/138 (76.1%). All 24 baseline and 23 rules handoffs remain in the denominator. Joint-program cover-call error rates are 5.3% and 5.0%, respectively; all shape conditions have zero underfill. Overflow rates expose the resource demands of retaining real calls under the frozen training template.

TABLE S31. E9 DISPATCH VIEW: MAXIMUM BA (%) WITH 95% INTERVAL AND MAXIMUM AUROC ACROSS THE FIXED SUITE, ON 138 TEST EPISODES PER PROMPT. BA AND AUROC ARE SEPARATE SUITE MAXIMA.

Prompt	Condition	Cancel BA [CI]	AUC	Return BA [CI]	AUC
baseline	Raw	97.2 [90.0, 100.0]	0.997	97.0 [88.9, 100.0]	0.972
baseline	Padding only	97.2 [90.0, 100.0]	0.997	97.0 [88.9, 100.0]	0.972
baseline	Shape only	55.6 [50.0, 63.0]	0.556	57.7 [50.0, 67.4]	0.620
baseline	Joint	55.6 [50.0, 63.0]	0.556	57.7 [50.0, 67.4]	0.620
rules	Raw	97.2 [90.0, 100.0]	0.972	97.0 [88.9, 100.0]	0.970
rules	Padding only	97.2 [90.0, 100.0]	0.972	97.0 [88.9, 100.0]	0.970
rules	Shape only	56.5 [50.0, 68.3]	0.655	64.7 [54.3, 77.4]	0.693
rules	Joint	56.5 [50.0, 68.3]	0.655	64.7 [54.3, 77.4]	0.693

TABLE S32. E9 NETWORK VIEW: MAXIMUM BA (%) WITH 95% INTERVAL AND MAXIMUM AUROC ACROSS THE FIXED SUITE, ON 138 TEST EPISODES PER PROMPT. BA AND AUROC ARE SEPARATE SUITE MAXIMA.

Prompt	Condition	Cancel BA [CI]	AUC	Return BA [CI]	AUC
baseline	Raw	97.2 [90.0, 100.0]	0.993	94.0 [85.8, 100.0]	0.948
baseline	Padding only	66.1 [50.0, 83.9]	0.634	55.5 [50.0, 73.0]	0.567
baseline	Shape only	60.8 [55.4, 71.3]	0.657	51.9 [50.5, 57.8]	0.579
baseline	Joint	50.0 [50.0, 60.6]	0.500	50.0 [50.0, 65.5]	0.500
rules	Raw	97.2 [90.0, 100.0]	0.978	93.5 [85.0, 99.5]	0.937
rules	Padding only	67.6 [50.0, 85.0]	0.636	57.9 [50.0, 78.9]	0.548
rules	Shape only	52.8 [50.0, 65.5]	0.588	53.2 [50.4, 63.9]	0.544
rules	Joint	50.0 [50.0, 65.1]	0.500	60.6 [50.0, 73.3]	0.641

S14.5. Interpretation and reproducibility

The completed cohort shows strong raw-trace intent inference and a large reduction under joint transport intervention. Rules improve mechanical adherence to the two read-order rules while leaving network BA above 93%. Baseline joint shaping reaches 50% BA for both targets, reducing the raw scores by 47.2 and 44.0 points. The finite-suite criterion requires the BA interval upper bound below 60% and point AUROC below 0.65. Baseline shape-only return meets it for that target; no condition meets it for both targets. Baseline joint upper bounds are 60.6%/65.5%, leaving the absolute-threshold conclusion inconclusive. The full-sample suite maximum equals the 50% majority baseline; several fitted attacks make heterogeneous predictions whose customer-resampled BA can exceed 50%. With only five cancellation-positive and seven return-positive customers, reselecting the maximum within each draw yields wider intervals. This mechanism does not imply a universal 10-point resolution floor or prove absence of residual population signal. The paired reduction intervals above exclude zero. Supplement S16 adds a task-permutation reference; the primary intervals remain clustered by customer. Rules-plus-joint return BA is 60.6%; its rules-minus-baseline contrast includes zero, as does the reward contrast.

The evidence covers one public retail environment and one mutable model alias, which also supplies user simulation. All 115 tasks were available during prior project development. The formal customer holdout therefore provides internal evaluation; external independent replication, broader model/environment transfer, provider-traffic measurement and interactive feedback effects remain follow-up questions.

The full cohort contains 11,586 model decisions, 2,760 TLS sessions and 176 attack-prediction sets. The frozen verifier passed after training-only recalibration, sandbox state replay, cover reconstruction, TLS accounting, metric/interval recomputation and request-provenance checks. Additional checks bind all agent actions and dialogue to actual model outputs and re-score required response strings. Offline attacker refitting reproduces the saved results and predictions exactly.

`experiments/sp2027-r4-full-20261023/` supplies full results, predictions and captures, plus the 27 new episodes and model records needed to extend the immutable 663-run snapshot. The three separately reported snapshot test episodes join the 27 new ones to expand the primary test from 246 to 276. `reproduce.py --reanalyze` reconstructs the full cohort from the base archives and delta, reruns analysis and executes the original frozen verifier without invoking a model. `verify_extension.py` checks decision alignment and provenance. The earlier snapshot, historical R2/R3 studies and pilot archives retain their original denominators and bytes.

TABLE S33. HELD-OUT UTILITY AND OVERHEAD. REWARD AND SESSION OVERFLOW RATES ARE PERCENTAGES; RATIOS USE RAW TRANSPORT WITHIN THE SAME PROMPT.

Prompt	Condition	Reward [CI]	Calls×	Bytes×	Count overflow	Cap overflow	Underfill
baseline	Raw	79.7 [69.0, 89.7]	1.00	1.00	0.0	0.0	0.0
baseline	Padding only	79.7 [69.0, 89.7]	1.00	4.02	0.0	6.5	0.0
baseline	Shape only	79.7 [69.0, 89.7]	3.83	4.16	38.4	0.0	0.0
baseline	Joint	79.7 [69.0, 89.7]	3.83	15.39	38.4	39.9	0.0
rules	Raw	76.1 [63.3, 88.7]	1.00	1.00	0.0	0.0	0.0
rules	Padding only	76.1 [63.3, 88.7]	1.00	4.04	0.0	6.5	0.0
rules	Shape only	76.1 [63.3, 88.7]	3.88	4.27	44.2	0.0	0.0
rules	Joint	76.1 [63.3, 88.7]	3.88	15.72	44.2	38.4	0.0

TABLE S34. MECHANICAL ADHERENCE TO THE TWO PROMPT-SPECIFIED READ-ORDER RULES (% WITH CUSTOMER-CLUSTER INTERVAL).

Prompt	One successful customer read first	Order read before write	Both
baseline	86.2 [74.5, 97.3]	100.0 [100.0, 100.0]	86.2 [74.5, 97.3]
rules	97.1 [91.7, 100.0]	100.0 [100.0, 100.0]	97.1 [91.7, 100.0]

TABLE S35. PAIRED PROMPT CONTRASTS USING COMMON CUSTOMER-BOOTSTRAP DRAWS.

Rules minus baseline	Difference [95% CI], pp
Task reward	-3.6 [-9.5, 1.4]
Raw network, cancel BA	0.0 [-8.3, 8.3]
Raw network, return BA	-0.5 [-2.6, 0.0]
Joint network, cancel BA	0.0 [-1.7, 7.4]
Joint network, return BA	10.6 [-13.3, 22.8]

TABLE S36. POST HOC PAIRED PROGRAM CONTRASTS. THE FIXED ATTACK SUITE IS RE-MAXIMIZED UNDER EACH CONDITION WITHIN EACH COMMON CUSTOMER-BOOTSTRAP DRAW. ORIGINAL PREDICTIONS AND THE PREDECLARED ABSOLUTE CRITERION ARE UNCHANGED.

Prompt	Target	Joint minus raw BA [95% CI], pp
baseline	cancel	-47.2 [-50.0, -33.3]
baseline	return	-44.0 [-48.9, -27.8]
rules	cancel	-47.2 [-50.0, -26.0]
rules	return	-33.0 [-44.3, -18.9]

S15. Cross-Fitting, Operating Points and Leakage Localization (E10–E12)

E10 retrospectively reuses the E9 defense design and attack suite under customer-grouped stratified 5-fold cross-fitting over all 690 completed episodes (115 tasks, 53 customers; 14 cancellation-positive and 18 return-positive customers). In each fold, the count template, donor pools, size caps and all attackers are fit only on the other folds. Out-of-fold predictions are pooled, and intervals use 2,000 customer-cluster bootstrap draws with the suite maximum recomputed per draw. The six analyzed configurations are specified in `r5lib.py`. E11 selects among them using inner customer folds of the 69 frozen training tasks only (rule in `e11_select.py`), then evaluates it on the 46 original test tasks. These analyses reuse an already analyzed cohort; only the original E9 protocol was frozen before test collection. E12 ablates undefended traces to locate the signal.

Records are computed with an analytic TLS 1.2 AES-GCM model (fragment plus 24 bytes, 16,384-byte fragments). The model is validated against all 2,760 socket captures of the completed cohort and 120 fresh loopback captures of new cross-fitted sessions. `verify_r5.py` also checks byte-identical reproduction of the E9 calibration, all sessions, and all 32 result cells of both the 660-run primary grid of the 663-run snapshot and the completed cohort. The same analysis on the 660-episode snapshot grid is retained in `snapshot-660/`; there the bucket point gives 50.6%/51.6% with upper bounds of 59.2%/58.3%. Cross-fitted templates use more training customers than E9 and are therefore larger, which raises call overhead (5.0× versus 3.8×). A quantile template is a natural next operating point.

TABLE S37. E10 CROSS-FITTED NETWORK VIEW: ALL CONFIGURATIONS AND PROMPTS. CORE = R4 SUITE; EXT. ADDS RANDOM FOREST, k-NN AND A FOREST SIZE DECODER. SA-AUC: SIGN-AGNOSTIC AUROC $\max(A, 1 - A)$.

Configuration	Prompt	Cancel core [CI]	Ext.	SA-AUC	Return core [CI]	Ext.	SA-AUC	Bytes×/Calls×
R4 point (all tools, constant) (padding only)	baseline	68.1 [55.4, 78.9]	68.1	0.63	67.1 [57.2, 77.0]	67.1	0.63	3.9/1.0
R4 point (all tools, constant)	baseline	60.5 [51.5, 70.5]	60.5	0.59	52.1 [50.0, 60.3]	52.7	0.63	19.8/5.0
R4 point (all tools, constant) (padding only)	rules	64.4 [54.6, 75.8]	64.4	0.60	63.3 [52.7, 74.2]	63.3	0.63	3.9/1.0
R4 point (all tools, constant)	rules	58.8 [50.0, 68.6]	58.8	0.57	59.4 [50.5, 68.0]	59.4	0.58	19.9/5.0
All tools, 1024-B buckets	baseline	52.2 [50.0, 60.5]	52.2	0.54	52.7 [50.0, 60.2]	52.7	0.61	12.4/5.0
All tools, 1024-B buckets	rules	53.4 [50.0, 60.6]	53.4	0.57	50.7 [50.0, 57.3]	52.4	0.56	12.4/5.0
All tools, constant, q80 cap	baseline	56.5 [50.5, 66.6]	56.5	0.55	60.2 [55.0, 66.0]	60.2	0.61	15.2/5.0
All tools, constant, q80 cap	rules	51.0 [50.0, 57.8]	51.0	0.54	51.3 [50.0, 56.4]	51.3	0.61	15.3/5.0
Writes only, constant	baseline	53.5 [50.0, 65.8]	56.1	0.60	56.1 [50.0, 67.4]	56.1	0.63	11.3/2.9
Writes only, constant	rules	62.6 [50.0, 74.4]	62.6	0.60	57.6 [50.0, 68.2]	57.6	0.58	11.4/2.9
Writes only, constant, q80 cap	baseline	53.7 [50.0, 66.4]	53.7	0.55	62.0 [51.2, 73.3]	62.0	0.66	8.6/2.9
Writes only, constant, q80 cap	rules	51.7 [50.0, 63.3]	53.4	0.58	58.6 [50.0, 70.3]	58.6	0.64	8.7/2.9
Writes only, 1024-B buckets	baseline	56.4 [50.0, 66.5]	56.4	0.55	58.1 [50.1, 68.6]	59.7	0.66	6.9/2.9
Writes only, 1024-B buckets	rules	52.6 [50.0, 63.8]	52.6	0.60	58.7 [51.6, 69.6]	58.7	0.59	7.0/2.9

TABLE S38. E11 INNER (TRAINING-FOLD) SELECTION STATISTICS. ELIGIBLE: ALL FOUR INNER NETWORK BAS $\leq 60\%$.

Configuration	C/base	R/base	C/rules	R/rules	Bytes×	Eligible
R4 point (all tools, constant)	54.9	53.5	58.4	52.6	15.2	yes
All tools, 1024-B buckets	53.3	58.1	55.3	56.8	9.8	yes
All tools, constant, q80 cap	56.8	56.8	53.1	54.8	11.7	yes
Writes only, constant	67.1	50.0	77.9	57.2	9.4	no
Writes only, constant, q80 cap	71.1	64.9	67.5	71.6	7.1	no
Writes only, 1024-B buckets	56.7	67.4	53.7	64.8	5.9	no

Rules-prompt homogenization: distinct read-only call sequences fall from 74 to 62 (entropy 5.63 to 5.37 bits), whereas state-changing sequences stay at 37 versus 36 (4.08 versus 4.09 bits).

TABLE S39. E12 OBSERVER ABLATION ON UNDEFENDED TRACES, CROSS-FITTED (CORE-SUITE BA, %).

Observer	Prompt	View	Cancel	Return
full	baseline	dispatch	91.7	93.8
full	baseline	network	91.7	85.5
full	rules	dispatch	90.7	93.8
full	rules	network	90.7	85.5
no writes	baseline	dispatch	53.1	65.7
no writes	baseline	network	55.2	59.9
no writes	rules	dispatch	60.8	66.2
no writes	rules	network	55.2	59.6
writes only	baseline	dispatch	91.7	93.8
writes only	baseline	network	91.7	84.9
writes only	rules	dispatch	90.7	93.8
writes only	rules	network	90.7	84.1

S16. Conditional Task-Permutation Reference

This post hoc diagnostic holds the fitted E9 attackers fixed and permutes labels over the 46 test tasks 2,000 times (seed 20261023), keeping each task's three repetitions together. Each draw recomputes the attack-suite maximum. It characterizes a task-label exchangeability reference; it does not preserve dependencies among tasks belonging to one customer. Consequently, the tail fractions below are conditional reference values, not customer-valid population p-values or evidence of equivalence. The original customer-cluster intervals, paired contrasts and frozen absolute criterion remain the primary assessments.

TABLE S40. ALL 32 E9 CELLS: OBSERVED MAXIMUM BA, PERMUTATION MEAN AND 95TH PERCENTILE (%), CONDITIONAL TAIL FRACTION $(1 + \#\{T_b \geq T_{\mathrm{obs}}\})/2001$, AND POSITIVE EXCESS OVER THE 95TH PERCENTILE (PERCENTAGE POINTS).

Prompt	View	Condition	Target	BA	Ref. mean	Ref. 95th	Tail	Excess
baseline	dispatch	Padding	cancel	97.2	54.6	65.3	0.0005	31.9
baseline	network	Padding	cancel	66.1	55.2	70.8	0.0755	0.0
baseline	dispatch	Padding	return	97.0	54.1	65.1	0.0005	31.9
baseline	network	Padding	return	55.5	56.2	67.4	0.4883	0.0
baseline	dispatch	Raw	cancel	97.2	54.6	65.3	0.0005	31.9
baseline	network	Raw	cancel	97.2	55.5	68.5	0.0005	28.7
baseline	dispatch	Raw	return	97.0	54.1	65.1	0.0005	31.9
baseline	network	Raw	return	94.0	55.3	65.4	0.0005	28.7
baseline	dispatch	Joint	cancel	55.6	54.2	62.8	0.3048	0.0
baseline	network	Joint	cancel	50.0	54.7	66.2	1.0000	0.0
baseline	dispatch	Joint	return	57.7	53.0	58.7	0.1034	0.0
baseline	network	Joint	return	50.0	54.1	62.6	1.0000	0.0
baseline	dispatch	Shape	cancel	55.6	54.2	62.8	0.3048	0.0
baseline	network	Shape	cancel	60.8	54.9	63.3	0.0870	0.0
baseline	dispatch	Shape	return	57.7	53.0	58.7	0.1034	0.0
baseline	network	Shape	return	51.9	54.3	60.4	0.7956	0.0
rules	dispatch	Padding	cancel	97.2	54.6	68.5	0.0005	28.7
rules	network	Padding	cancel	67.6	55.7	69.2	0.1034	0.0
rules	dispatch	Padding	return	97.0	54.4	65.1	0.0005	31.9
rules	network	Padding	return	57.9	55.6	67.2	0.3033	0.0
rules	dispatch	Raw	cancel	97.2	54.6	68.5	0.0005	28.7
rules	network	Raw	cancel	97.2	55.1	68.5	0.0005	28.7
rules	dispatch	Raw	return	97.0	54.4	65.1	0.0005	31.9
rules	network	Raw	return	93.5	55.6	65.8	0.0005	27.8
rules	dispatch	Joint	cancel	56.5	53.2	59.7	0.1889	0.0
rules	network	Joint	cancel	50.0	53.8	63.2	1.0000	0.0
rules	dispatch	Joint	return	64.7	53.5	60.7	0.0115	4.0
rules	network	Joint	return	60.6	54.1	60.6	0.0680	0.0
rules	dispatch	Shape	cancel	56.5	53.2	59.7	0.1889	0.0
rules	network	Shape	cancel	52.8	54.3	60.8	0.6122	0.0
rules	dispatch	Shape	return	64.7	53.5	60.7	0.0115	4.0
rules	network	Shape	return	53.2	54.6	59.6	0.7081	0.0

The baseline raw network cells exceed the reference 95th percentiles (68.5%/65.4%), whereas baseline joint cells have no positive excess. This is consistent with the measured paired reductions; failure to exceed a reference threshold is not a population guarantee. The diagnostic uses saved predictions and makes its task-level exchangeability assumption explicit.

S17. Shadow Cover: Deployment Paths and Observer Boundaries

This section translates the measured isolation mechanism into deployment contracts. The evidence is the existing E6/E7 reference-program study and E9–E12 real-agent trajectory analysis. The three paths below are engineering designs grounded in existing API primitives; they have not been evaluated as production integrations.

S17.1. What padding leaves visible, and what a tunnel changes

E7's no-cover bucketed framing leaves return BA at 73.9%. E9's baseline per-call padding leaves cancellation/return BA at 66.1%/55.5%. A padded request and response still contribute observable records: invocation count, direction changes and ordered turns remain available even if individual sizes coincide. The joint E9 intervention reduces the two measured BAs to 50.0% by combining call-profile shaping with constant framing. This identifies a useful intervention boundary; it does not prove that executing dummy tools is necessary for every network threat model.

RFC 9347 describes a different construction: aggregate and fragment inner traffic into fixed-size ESP packets, send at a constant rate and use padding when no inner data is ready [36]. For a passive observer of that tunnel alone, dummy transport traffic can conceal application call boundaries without reaching a tool. Protecting workflow duration also requires a common or otherwise workload-independent transmission horizon in both directions; destination choice, tunnel start/stop and congestion responses remain part of the observation contract. Our per-call framing ablations are not measurements of this complete construction, and provide no comparative cost or security ranking against it.

At the orchestrator-to-tool boundary, a multiparty relay may instead see decoded RPCs, tool names or forwarding events after a transport tunnel terminates. A tunnel's padding disappears at that point. Cover must survive to the relevant observer as plausible calls, or application fields must remain encrypted through the relay. E6/E9 evaluate the specified call-name/length and network projections. They do not establish indistinguishability against unrestricted argument inspection, routing logs or a tool provider's state view.

S17.2. Three execution paths

1. Authorized reads and verified no-ops. Where authorized read operations suffice for the protected observation, issue them against permitted live resources and discard their responses before the planner. This can avoid a replicated backend. The read-only decoy pattern is related to WRA [37]; its ability to conceal write intent depends on what the observer sees. A receiver that recognizes write methods cannot be confused by adding only reads. The E12 writes-only ablation directly motivates covering the write-bearing part of the profile as well.

A no-op needs the invariant $f(s, a) = s$ for the designated protected state. Idempotence alone permits the first call to change state. Reads can also incur billing, access logs, cache changes or rate limits, so the adapter must specify its state scope and budget those effects. Cover selection must respect resource authorization and avoid exposing unrelated users' data. No claim about reads being the majority of production tool traffic is needed for this path.

2. Native noncommit and isolated test environments. Kubernetes server-side dry-run performs admission and validation without persisting the resource; unsafe admission side effects cause rejection, and authorization matches the corresponding real request [38]. This supplies a concrete noncommit primitive. Stripe sandboxes instead keep a separate test environment in which payments do not move real money [39]. They instantiate isolated state, rather than a dry-run flag on a live payment. Neither feature alone establishes cover indistinguishability: an observer that reads the dry-run flag, test credential or separate destination can classify the call.

Database transactions can execute operations and then roll back their transactional updates [40]. This path is suitable only when relevant effects lie within the rollback contract. PostgreSQL sequence updates, for example, are not undone by transaction abort [41]; locks, quotas and external effects also need explicit treatment. A rollback is not a general substitute for isolating irreversible remote actions. Common ingress framing must hide mode metadata from the intended observer, while the provider performing the operation remains trusted for that distinction.

3. A trusted gateway with isolated shadow endpoints. A tool/MCP gateway can receive a common envelope containing an authenticated, encrypted real/cover marker. A fixed-length marker field is present in both modes and is readable only by the trusted gateway. It forwards real calls unchanged, routes cover writes to an isolated simulator and suppresses cover responses to the planner. The simulator can maintain only the state needed by supported operations; it need not clone an entire enterprise service. Error handling must fail closed: an unsupported shadow operation must never fall through to a live write.

This gives cover calls real local execution and real response packets, but does not automatically reproduce production latency or size distributions. A shared response scheduler and framing policy must normalize the exposed differences; their delay and resource costs need measurement. Retries, authentication failures, quota exhaustion and overflow must follow the same isolation contract. An encrypted marker alone is insufficient if a relay can infer the branch from destinations or correlate ingress calls with gateway-to-provider traffic.

S17.3. Where the protection ends

The intended topology is an orchestrator, an optional forwarding relay, a trusted gateway, and separate live and shadow executors. Table S41 distinguishes roles that are often conflated as a "proxy." The trust requirement follows

TABLE S41. DEPLOYMENT CONTRACTS BY OBSERVER. THESE ARE ARCHITECTURAL CONDITIONS, NOT NEWLY MEASURED GUARANTEES.

Observer	Required protection boundary
Passive ingress link	Joint call/size shaping or a complete transport tunnel; include timing and session horizon in the threat model.
Call-visible relay	Cover must reach it; marker, mode, routing and identifying arguments must be hidden or matched. E9 covers only its specified projection.
Trusted gateway	Reads the marker and routes branches; it is outside the protected observer set.
Downstream link	Ingress-only cover is insufficient if only real calls traverse this link; add protection on that link.
Tool provider	Sees live effects and mode/credentials; hiding committed intent requires additional assumptions beyond this design.

the ability to distinguish or commit the two branches, rather than organizational ownership alone.

For a passive client-to-gateway observer, gateway-local cover can supply the measured kind of call-profile intervention. For an honest-but-curious relay, a protection claim additionally requires that its complete view exclude the discriminator and reveal no reliable substitute through arguments, timing or routing. A TLS-terminating relay that can read the marker does not satisfy this condition. The trusted gateway and the provider's commit log are not protected by moving cover into a local simulator.

All three paths preserve the logical planner transcript only when real observations pass unchanged and cover cannot affect live state or planner-visible feedback. That is the invariant checked in the existing replay. A deployment evaluation would additionally measure interactive reward, latency, failures and quotas, refit attackers on each exposed projection, and compare a full tunnel at matched workload and cost. No such production or tunnel comparison is claimed here. The contribution of shadow cover is a concrete way to shape an observable call profile while isolating its state effects, with an explicit choice of where that isolation must be enforced.

Complete indexed-family citation inventory. The following citation groups enumerate all 82 indexed families. Categories are core eligibility and origin, not mutually exclusive threat objectives. The machine-readable index binds each key to its source version and locators. Core (58 families): [31], [42], [37], [43], [32], [44], [45], [46], [35], [47], [48], [49], [50], [34], [51], [52], [53], [33], [54], [23], [2], [55], [56], [3], [4], [57], [5], [12], [58], [8], [9], [6], [27], [7], [10], [59], [11], [60], [13], [14], [61], [62], [63], [64], [65], [66], [67], [68], [1], [69], [70], [71], [72], [73], [74], [75], [76], [77].

Adjacent (24 families): [15], [78], [79], [16], [17], [18], [19], [20], [21], [22], [80], [81], [82], [28], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92].

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